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Mentor di Logica

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a cura di
Luisella Caire, Paola Suria Arnaldi

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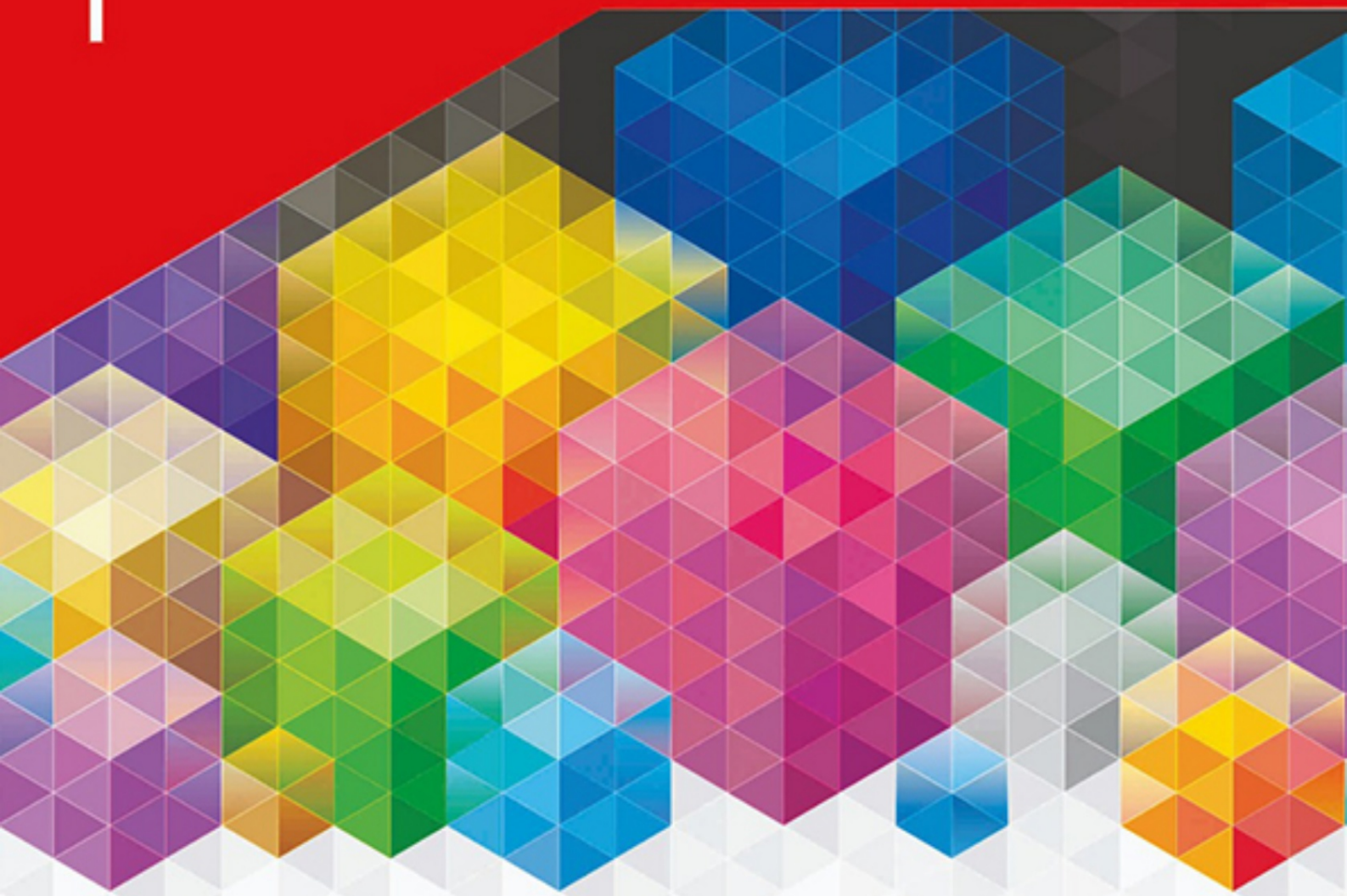

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Logic Mentor

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Presentation

This text, whether in print or (possibly) in the form of a PDF file, must be used as an interactive book, either by leafing through it by hand following the instructions provided along the way, or by following the links in the PDF file; practically all PDF viewers for calculators, iPads of various sizes, and tablets that are not too small, seem to be able to navigate a PDF document by following the hyperlinks.

This booklet is a self-learning tool; it is not intended to simulate a test, but to practice answering closed-ended questions on specific subjects that are generally included in university entrance tests.

The booklet has a particular shape and cannot be used by reading the pages sequentially; a sequential reading is intentionally almost incomprehensible. The reading must be done in this way: you start from the first question; you read the text and choose an answer by going to the page indicated at the end of the answer. If the answer is correct, the new page confirms it, perhaps adding some small information and indicates the page where to find the next question. If instead the chosen answer is wrong, the new page explains why it is wrong and at the bottom indicates the page to return to the question for which the wrong answer was chosen.

As you can see, the reading path is forced by the indications of the pages to move to. Instead, sequential reading is practically impossible because both the questions and the answers are distributed "randomly" along the text and there are no two logically consecutive elements (questions or explanations to the answers) placed in the text in adjacent positions.

In this particular mentor, questions of propositional and graphic logic are presented; sometimes it is necessary to do some simple calculations (which can be developed mentally), but it is emphasized that these questions that require simple calculations do not require any specific knowledge of mathematics that has not been known since compulsory school.

Logic as such is not generally the subject of any course in secondary schools; but the questions are formulated without using logical "jargon", but in natural language; therefore no specific preparation is necessary; instead it is necessary to know how to connect sentences or drawings intelligently without confusing causes and effects.

The time to be devoted to each question should not be too long; however, reviewing or self-learning logical concepts is a very different thing from taking an admission test carried out through a questionnaire full of closed multiple-choice questions, having an average of a couple of minutes available for each question. The time to respond in this type of test together with the difficulty of the question itself is an essential element for the effectiveness of the admission test; it allows to better separate the sample of candidates who appear for the test in order to discriminate the preparation of individual candidates. Usually this does not prevent brilliant candidates, who have followed secondary studies poorly suited to the chosen faculty, from showing their abilities; equally well it is possible to better identify those candidates who, despite having followed secondary studies suited to the faculty, are not so interested in the disciplines that form the basis of their future studies.

This is why reviewing your logical reasoning skills is important; just as it is important to discover before tackling university studies that you do not have a logical way of reasoning in some areas, so that you can make the most of the explanations given for each answer to each question in this text.

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How to use the mentor

In the mentor, the questions and the explanations of the closed answers they contain are mixed randomly, but next to each answer, the page on which the explanation is located is indicated by a black hand whose index finger points to the page to which to move; on the indicated page, the explanation is clearly identified as answer 'X' to question 'Y'. The explanation consists of a title that classifies the answer as right or wrong and is followed by a short explanatory text possibly accompanied by formulas and drawings. At the end of the short explanatory text, another black hand indicates which page to go to in order to continue using the mentor; in particular, if the answer given was wrong, the hand indicates to return to the question to which you were unable to answer, while if the answer was right, the hand indicates the page of the next question.

Note that the questions are numbered sequentially in the mentor, but this game of referrals via page pointers proceeds in a random order, which however at the end of the reading will have allowed you to reach each question.

Reading does not occur sequentially according to the page numbering, but only following the sequence of pages indicated by the various hands. Whether the mentor is printed or in the form of an e-book to be read on the screen, reading according to the natural sequence of pages is virtually impossible. In short, you are forced to think carefully about each question to proceed with the sequence of correct answers.

If a question is answered incorrectly once or twice, it is likely that the topic on which the question is based constitutes a gap or that the topic has not been assimilated properly. The mentor's aim of highlighting the deficiencies in logical reasoning is achieved.

The reader then tries to understand why the answers given were wrong; explanations usually clarify what errors in logical reasoning were made, so understanding the errors also helps you understand how to reason logically.

If, on the other hand, the mentor's guided reading is followed well, it means that the reader has the necessary knowledge and skills; it will be his responsibility, if necessary, to point out if he has encountered questions that have forced him to do a lot of mental and material work to repeat the logical reasoning carried out and he will be able to acquire greater operational "ease". Once again, the mentor's aim is achieved; the reader has in fact identified his possible weaknesses and has the opportunity to practice to strengthen himself in the topics that have been more difficult for him.

We hope that the readers of this text will find the method followed in this mentor useful; we know from experience that a book from about fifty years ago, set up in the same way as this mentor, was very useful to the young people of that time. Today there are more modern means, but learning is still based on individual cognitive experiences, exactly as it was then.

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Start

Start from the page containing the question from which you want to begin reading the mentor, as indicated in the box below.

question 41 on page 67

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Question 1

A company delivered frozen food to your home and now you have to pay the delivery man (or delivery men) who actually carried out the task. The delivery was made with a van.
That day only the employees Aldo, Bruno, Ciro were available. Knowing that Ciro was not works if Aldo doesn't work too and Bruno doesn't know how to drive, then:

A. Aldo will be paid B.	ÿ page 76
only Bruno will be paid C.	ÿ page 44
neither Aldo nor Bruno will be paid D. Aldo	ÿ page 12
will not be paid and Bruno will be paid E.	ÿ page 65
Ciro will be paid and Aldo will not	ÿ page 47

Question 35 - Answer C is wrong

It is not the exact square; the law that generates the numbers on the sides and vertices cannot be identified, internally or externally to the figure.

ÿ to question 35 on page 55

Question 46 - Answer D is wrong

What if there were 5? Try a counter example.

ÿ to question 46 on page 73

Question 32 - Answer C is wrong

Some trustworthy people are involved in politics!

ÿ to question 32 on page 52

Question 33 - Answer C is wrong

Let ÿ be the set of prime numbers, say that

ÿ ÿ ÿ ÿ > ÿ ÿ ÿ e + 2 ÿ ÿ

it is not the refutation of the conjecture.
Denial of ownership requires that

ÿ ÿ ÿ ÿ ÿ > with ÿ ÿ, + 2 ÿ ÿ

ÿ to question 33 on page 53

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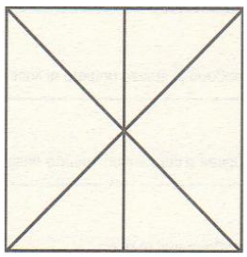
Question 2

How many triangles can you identify in the following figure?



A. 6
B. 12
C. 10
D. 8
E. 16

ÿ page 72 ÿ page 70 ÿ page 44 ÿ page 36 ÿ page 25

Question 27 - Answer C is wrong

You can play ball before 1pm and after 4pm on non-holidays, but it is not mandatory!
The formulation error is generated by the position shift of the logical connective NOT.
Indeed:

it is allowed to play… means it is not forbidden to play…

Instead

it is forbidden NOT to play… it means you must, it is mandatory to play…!

ÿ to question 27 on page 43

Question 35 - Answer B is wrong

The missing figure is not a hexagon; moreover, it is not possible to find a law that links the writing of numbers on the sides and vertices, internally or externally.

ÿ to question 35 on page 55

Question 9 - Answer E is correct

Luigina herself says:

yesterday (Wednesday) I DID NOT take a bath and I went to the market

ÿ to question 22 on page 37

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Question 3

A chemist, studying a solution that had turned orange, found that in it
was sodium or potassium (or both) present; he also observed that if there was NO sodium,
there was iron, and that, if there was potassium, there was also iodine.
Which of these situations can occur?

A. The solution contains only potassium and iron ÿ page 35

B. The solution contains only iron and iodine ÿ page 36

C. the solution contains sodium and potassium, and does not contain iodine ÿ page 45

D. The solution contains neither sodium nor iodine ÿ page 25

E. The solution contains only sodium ÿ page 37

Question 37 - Answer D is wrong

If the budget was not cut, the necessary and sufficient condition for prices to remain
stable was that all taxes should increase and not just those of state employees.

ÿ to question 37 on page 59

Question 24 - Answer E is correct

We show that, for any table of the proposed type, ÿ .

- if and are on the same line, then < of the numbers , because, by definition, it is the smallest on their own line;
- if and are in the same column, then < larger than the , because by definition, it is the most numbers in its column;
- if and are neither on the same row nor on the same column, then calling the number that is on the same row as and in the same column as , by the relationship must be valid < < definition of and .

Therefore < .
The special case

(1 3 2 4) ÿ

1	2				
1 1	3 3 3			2 4 3	

ÿ 3 = =

prove that it can also be = ; therefore ÿ .
In this question, one could also proceed by exclusion, after having analysed the other four
answers and having demonstrated that they are false, confident that every well-formulated question has only one answer
correct answer!

ÿ to question 19 on page 34

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Question 4

With which pair of numbers would you continue the following sequence of integers?

1, 2; 2, 4; 5, 8; 10, 14; 17, 22; 26, 32; … , …

A. 40, 80
B. 46, 64
C. 40, 67
D. 37, 44
E. 44, 16

ÿ page 69 ÿ page 70 ÿ page 72 ÿ page 18 ÿ page 48

Question 18 - Answer D is correct

If there are two glasses on the table then it is certainly true that there is a glass on the table! Attention, the meaning of a sentence, in logic, does not necessarily correspond to the interpretation that can be given to that sentence in common language.

ÿ to question 5 on page 11

Question 15 - Answer D is wrong

Another is the right answer! If only there were only 5! Play to convince yourself!

ÿ to question 15 on page 27

Question 47 - Answer E is wrong

With two archaea, time is not halved! If two archaea are put into the pond, the situation present on the second day is created starting with only one archaea. ÿ to question 47 on page 74

Question 9 - Answer D is wrong

The day before yesterday was Tuesday and Luigina took a bath, so she went to the market!

ÿ to question 9 on page 15

Question 10 - Answer E is wrong

3 colors are too many.

ÿ to question 10 on page 20

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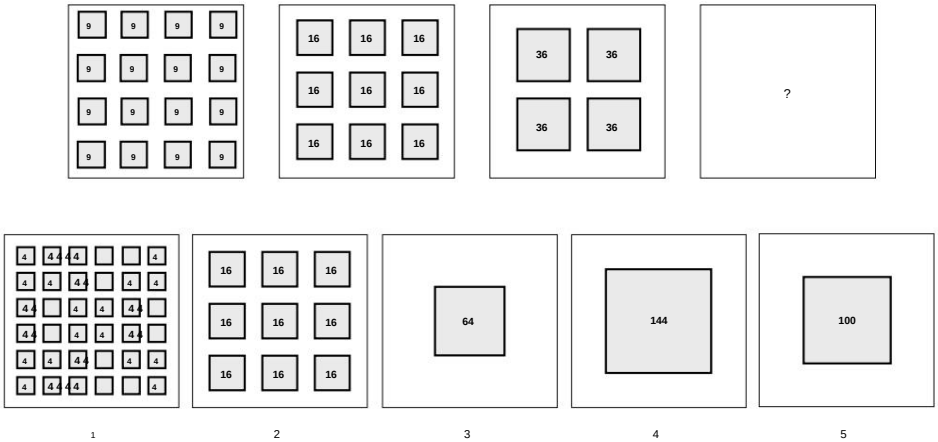
Cheese No. 1

Which figure completes the sequence?

Question 5

Which of the squares numbered 1 to 5 correctly replaces the box containing the question mark?



A. 1 A. The square 5 2 ÿ page 24

B. B. The square 3 3 C. ÿ page 23

C. The square 1 4 D. ÿ page 61

D. The square 2 5 E. ÿ page 26

AND. The square 4 ÿ page 19

Question 38 - Answer E is wrong

Julius Caesar won more than some others.

ÿ to question 38 on page 63

Question 50 - Answer B is wrong

The law of formation of the sequence of numbers has not been correctly identified.

ÿ to question 50 on page 77

Question 9 - Answer B is wrong

If Luigina takes a bath on Tuesday then she goes to the market. She doesn't say, but if she takes a bath every Tuesday!

ÿ to question 9 on page 15

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Question 6

Of the two brothers Romulus and Remus, one is always sincere and the other is always a liar. If Romulus says that the mother of the two is called Silvia and Remus affirms that she is blonde, it can be deduced with certainty that:

A. if the mother is blonde, her name is not Silvia B. if the mother is called Silvia, then she is blonde C. the mother's name is Silvia and she's not blonde D. if the mother's name is not Silvia, then the mother is not blonde E. the mother's name is not Silvia

ÿ page 54 ÿ page 77 ÿ page 74 ÿ page 46 ÿ page 64

Question 15 - Answer C is wrong

That's not the right answer: draw a chessboard and play to convince yourself!

ÿ to question 15 on page 27

Question 41 - Answer E is wrong

The hypothesis

It is not true that every parliamentarian is an honest and competent person

it's equivalent to saying that

It is false that all parliamentarians are honest and competent,

without excluding, however, that there may be honest and competent parliamentarians. Certainly there is at least one parliamentarian who does not possess at least one of these two qualities: honesty or competence.

ÿ to question 41 on page 67

Question 1 - Answer C is wrong

If Aldo and Bruno are not paid, then Ciro made the deliveries, but Ciro works only with Aldo's help...

ÿ to question 1 on page 7

Question 41 - Answer D is wrong

The premise does not take away the hope that some honest and competent person exists, runs for office and becomes a member of parliament!

ÿ to question 41 on page 67

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Question 7

Chef Giovanni observes that when cooking a roast without using a gas oven, the meat is either raw on the inside or burnt on the outside or both.

It follows that:

A. if the roast is well cooked inside it has been cooked in a gas oven ÿ page 17 B. if the roast is well cooked inside or is not burnt on the outside

it was cooked in the gas oven ÿ page 45

C. if the roast is well cooked inside and not burnt on the outside

it was cooked in the gas oven ÿ page 52 D. if the roast is raw inside it has not been cooked in the gas oven ÿ page 47 E. if the roast has been cooked in the electric oven the meat is raw

inside and burned outside ÿ page 33

Question 13 - Answer A is correct

The overall statement is false

Whoever breathes is alive, Piero does not breathe, therefore Piero is dead

In fact the premise is true

whoever breathes

is alive but if Piero is not breathing it does not necessarily mean that Piero is not alive, that he is dead. Piero could, at that moment, be holding his breath.

ÿ is not equivalent to (NOT) ÿ (NOT)

He who breathes is alive is not equivalent to He who does not breathe is not alive, that is, he is dead.

ÿ to question 20 on page 35

Question 42 - Answer C is correct

One of the two hypotheses of the question is: if there are few of you, you eat well.

So: • if we eat well Mr. Aquilotto CANNOT deduce how many of us are eating: few or many;

• if we eat badly Mr. Aquilotto can deduce that we are definitely NOT in a few.

If we interpret the question from the point of view of mathematical logic, placed =

there are few of us and = we eat well

so long as

ÿ is equivalent to (NOT) ÿ (NOT)

Mr. Aquilotto can deduce that: if you eat NOT well (that is, if you eat badly) then you are NOT few (that is, you are many) ÿ to question 45 on page 72

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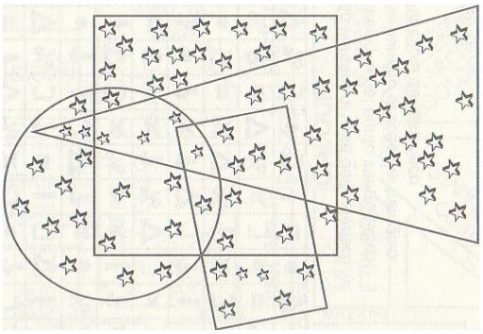
Question 8

Say how many stars are included in both the triangle and the circle but not in the square and not even in the rectangle.



A. 2 ÿ page 31
B. 3 ÿ page 41
C. 4 ÿ page 59
D. 5 ÿ page 24
E. 6 ÿ page 57

Question 50 - Answer C is correct

14 = 7 ÿ 2 + 0, 29 = 14 ÿ 2 + 1, 60 = 29 ÿ 2 + 2, 123 = 60 ÿ 2 + 3

Therefore:

= 123 ÿ 2 + 4 = 250, = 250 ÿ 2 + 5 = 505

Each number is double the one preceding it plus the number of its position in the sequence, decreased by two.
For example 14 is the second number in the sequence and is double the number that precedes it. (7) to which we must add 2 ÿ 2 = 0.
29 is the third number and double the number preceding it (14) and must be increased by 3 ÿ 2 = 1...

ÿ to question 32 on page 52

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Question 9

Luigina says:

- on Tuesday, if I take a bath then I go to the market. The day before yesterday was Tuesday, and I did the bathroom
- yesterday I didn't take a bath and I went to the market
- today I'll go to the market and maybe I'll even take a bath

It necessarily follows that:

A. every time Luigina goes to the market, she doesn't take a bath B. on Tuesdays Luigina always takes a bath C. if Luigina takes a bath on Wednesday then she doesn't go to the market D. the day before yesterday Luigina didn't go to the market E. sometimes Luigina goes to the market without having taken a bath

ÿ page 31 ÿ page 11 ÿ page 35 ÿ page 10 ÿ page 8

Question 14 - Answer B is wrong

The length of the stick, i.e. 90 cm, exceeds one of the three dimensions indicated in the standard 50 cm, 30 cm or 25 cm, therefore if you want to carry the stick in question with you you must pay an additional ticket. The information regarding the way in which the stick is held, i.e. horizontally, is completely superfluous.

ÿ to question 14 on page 26

Question 12 - Answer B is wrong

Let's look for a particular case that satisfies both the hypotheses of the question:

- # married people > # singles
- # males > # females

both when asked for the statement: • # couples > # single males

For example, let's choose two couples, one single male and zero single females. The number of couples is greater than the number of single males.

We have shown that the statement itself is definitely NOT False.

ÿ to question 12 on page 23

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Question 43 - Answer C is wrong

The King does not listen to the advice of the Grand Chamberlain to oppose the marriage only if the two young people intend to marry in Superga. The spouses can, therefore, also choose the beautiful Basilica of Superga to celebrate their wedding.

ÿ to question 43 on page 69

Question 12 - Answer E is wrong

Let's rewrite the hypotheses of the question:

- # married people > # singles
- # males > # females

and we add the response condition: • # single
males > # husbands

We look for a particular case in which all the assumptions are satisfied.
For example, 12 married people (6 husbands and 6 wives), 8 single males and 4 single females.
There are more single men than husbands, so the proposed case is possible and the statement is true. ÿ to

question 12 on page 23

Question 32 - Answer E is wrong

Only a few trustworthy people (and therefore only a few engineers) deal with politics. Therefore only a few engineers deal with politics.

ÿ to question 32 on page 52

Question 40 - Answer E is wrong

Mr. Rossi is left-handed, but not all left-handed people are stabbers, much less murderers.

ÿ to question 40 on page 65

Question 11 - Answer D is wrong

What relationships are there between the numbers 12 and 240 with the first or second sequence of numbers respectively?

ÿ to question 11 on page 22

Question 36 - Answer A is wrong

The correct answer is another: with only one 20 euro piece in the cash register it would have a different total.

ÿ to question 36 on page 56

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Question 11 - Answer C is correct

Let us identify the laws that generate the two sets of numbers, knowing that they may also be different.

Let's rewrite the numbers in the first row of the table:

1, 3, 6, 10, , 21, 28

The difference between a number and the one preceding it grows like the sequence of natural numbers:

ÿ 10 = 5 ÿ = 15; 21 ÿ = 6 ÿ = 15

Let's rewrite the numbers in the second row of the table:

1, 2, 6, 24, 120, , 5040

The ratio between a number and the one preceding it grows like the succession of natural numbers:

/120 = 6 ÿ = 720; 5040/ = 7 ÿ = 720

ÿ to question 47 on page 74

Question 7 - Answer A is wrong

The hypothesis

If you do NOT use a gas oven then the meat is
either raw inside or burnt on the outside or both

it means roast
well cooked inside and not burnt on the surface ÿgas oven
In fact, in mathematical logic:

ÿ is equivalent to (NOT) ÿ (NOT)

We know the roast is well done on the inside, but what about the outside?
The occurrence of only one of the two conditions does NOT guarantee that it was cooked in a gas oven ÿ

question 7 on page 13

Question 50 - Answer D is wrong

What is the relationship between the numbers 224 and 283 and the sequence?

ÿ to question 50 on page 77

Question 27 - Answer B is wrong

On Sundays you are NOT allowed to play ball in the yard, at any time of the day. ÿ to question 27 on page 43

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Question 21 - Answer D is wrong

At least one, but we are not sure if it is all seven, fears not being among the three most voted!

ÿ to question 21 on page 36

Question 34 - Answer C is wrong

Sorry, but the answer is not correct: it is easier to open the safe. Try to build the quaternions that respect the conditions of the question.

ÿ to question 34 on page 54

Question 40 - Answer B is wrong

Let's rewrite the hypotheses:

- if he stabbed the victim, then he is left-handed
- if he stabbed the victim, then he is the murderer

We can deduce that

if he stabbed the victim he is left-handed and he is the murderer

But the initial if does not ensure that the victim was stabbed. The murderer could have poisoned her…

ÿ to question 40 on page 65

Question 17 - Answer C is wrong

6 becomes 9 by adding one rod and removing one.

ÿ to question 17 on page 30

Question 4 - Answer D is correct

The difference between the first term of each pair and the corresponding term of the preceding pair grows as the sequence of odd numbers. If it is the first term of the pair:

(2 ÿ 1) = 1, (5 ÿ 2) = 3, … (ÿ 26) = 11 ÿ = 26 + 11 = 37

The difference between the second term of each pair and that of the previous pair grows as the sequence of even numbers increases.

(4 ÿ 2) = 2, (8 ÿ 4) = 4, … (ÿ 32) = 12 ÿ = 32 + 12 = 44

ÿ to question 15 on page 27

Question 46 - Answer A is wrong

This is not the right number of Cyrillic numbers; three or four examples will suffice.

ÿ to question 46 on page 73

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Question 5 - Answer E is correct

The number of gray squares inside each large square is a perfect decreasing square (16, 9, 4, ?). The only perfect square, smaller than 4, is 1. We therefore expect only one gray square. The numbers written in each square are perfect increasing squares (9, 16, 36, ?).
The only perfect square greater than 36, among the possible choices, is 144.
We can also see that the product of the number of squares by the number written on each square is always 144 (that is, the sum of all the numbers in each large square is always 144).

ÿ to question 28 on page 44

Question 19 - Answer E is wrong

Ruggiero cannot be behind the odd door 3, behind which hides a deadly dragon! ÿ to question 19 on page 34

Question 50 - Answer E is wrong

The law of formation of the sequence has not been identified.

ÿ to question 50 on page 77

Question 23 - Answer C is wrong

Agenore, who is not tone deaf, can listen to the blues and in that case, he does not win the lottery.
So this conclusion can be drawn from the premises; we are looking for a conclusion that is certainly false, and this is not.

ÿ to question 23 on page 38

Question 20 - Answer C is correct

Let's rewrite the hypotheses of the question:

- at least one male is married
- all graduates, if any, are not married
- all males are adults

We can deduce that: all
males are adults and at least one adult is married. ÿ to question 33 on page 53

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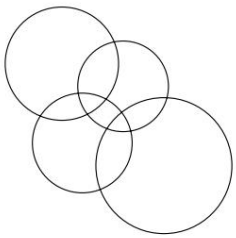
Question 10

We need to color the 11 regions bounded by the 4 circles in the figure so that no two regions that have a common arc are the same color.
How many colors do we need to use at least to satisfy this request?



A. 4
B. 6
C. 2
D. 5
E. 3

ÿ page 51 ÿ page 24 ÿ page 58 ÿ page 44 ÿ page 10

Question 48 - Answer B is wrong

The hypothesis of the question

We all like coffee, except Paola, who doesn't like it.

talks about both Paola and her friends, other than Paola.
The hypothesis is denied not only if Paola likes coffee, but also if, besides Paola, at least one other friend of Antonio does not like coffee.

ÿ to question 48 on page 75

Question 49 - Answer A is wrong

The denial of:

Umberto has at least one blond son

And

Umberto has no children or all of his children are NOT blond. The proposed answer is wrong because we know that Umberto's children, if they exist, are NOT blond and not that at least one of Umberto's children is NOT blond.

ÿ to question 49 on page 76

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Question 40 - Answer A is wrong

Let's rewrite the first hypothesis of the question:

if he stabbed the victim, then he is left-handed If Bianchi is not left-handed, he is certainly not the stabber; however, he could equally be the murderer. The victim could have been poisoned, strangled… by Bianchi.

We remind you that:

ÿ is equivalent to (NOT) ÿ (NOT)

NOT left-handed ÿNOT stabber

and

NOT stabber ÿ/ NOT murderer.

ÿ to question 40 on page 65

Question 48 - Answer C is wrong

If the premise of the question is

We all like coffee, except Paola, who doesn't like it.

then its negation is equivalent to:

Either Paola likes coffee or there is someone among her friends, besides Paola, who doesn't like coffee

The proposed answer

one of the friends, who is not Paola, doesn't like coffee

takes into consideration only one of the two eventualities, compatible with the denial of Antonio's affirmation.

ÿ to question 48 on page 75

Question 47 - Answer A is wrong

In 30 days the pond fills up if at the initial moment there is only one archaebacterium.

ÿ to question 47 on page 74

Question 31 - Answer E is correct

If with O, P, Q, R, S, we indicate the assets of Oronzo, Piero, Quirino, Rocco, Silvio, we can write the following inequalities:

$$O > R,\ S > R,\ S > P,\ P > Q,\ Q > O$$

If we compare the relationships we deduce that:

$$S > P > Q > O > R$$

So the third in order of wealth is Quirino.

ÿ to question 8 on page 14

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Question 11

Which of the numbers , should the proposals be included in the table?

1	3	6	10	1		21	28
2	6	24	120				5040

A. = 14 and = 720
B. = 14 e = 240
C. = 15 and = 720
D. = 12 e = 240
E. = 15 and = 240

ÿ page 29
ÿ page 37
ÿ page 17
ÿ page 16
ÿ page 31

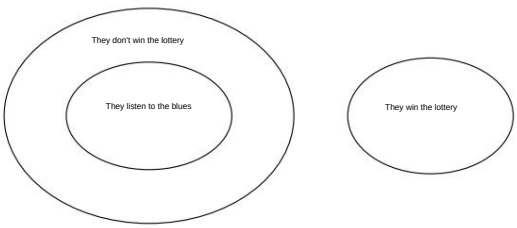
Question 14 - Answer D is wrong

The current bus regulation requires the purchase of an additional ticket if you are transporting a object that has at least one dimension greater than the three indicated.

ÿ to question 14 on page 26

Question 23 - Answer D is wrong

Let's represent the third hypothesis of the question using Venn diagrams:



It's true: all those who win the lottery do NOT listen to the blues!
With the logical implications we can reflect that

ÿ is equivalent to (NOT) ÿ (NOT)

So:

Those who listen to blues music do NOT win the lottery

ÿ

Lottery Winners DO NOT Listen to Blues Music

ÿ to question 23 on page 38

Machine Translated by Google

Question 12

A survey shows that in Italy there are more married people than singles and more males than females.

From this data we can deduce that only one of the following statements is definitely FALSE; which one?

A. In Italy there are more couples than single women ÿ page 25 ÿ page
B. In Italy there are more couples than single men C. 15 ÿ page 62 ÿ
In Italy there are more husbands than single women page 38 ÿ page
D. In Italy there are more single men than couples E. In 16
Italy there are more single men than husbands

Question 16 - Answer B is wrong

The different numbers that can be obtained from algebraic sums of the required type are not this.
Find a logical method that allows you to understand the type of numbers generated.

ÿ to question 16 on page 28

Question 5 - Answer B is wrong

The law that generates the number of squares has been understood, but the law that forms the numbers written on their faces has not been identified.

ÿ to question 5 on page 11

Question 32 - Answer A is wrong

There is a statement among those proposed that can be deduced from the hypotheses of the question, so it is not this one.

ÿ to question 32 on page 52

Question 17 - Answer E is wrong

5 becomes 6 by adding just one more rod to the display.

ÿ to question 17 on page 30

Question 22 - Answer B is wrong

Exactly 50% of the population over 55 years of age owns a television or refrigerator :

400 + 250 ÿ 150 = 500 ÿ 50%

ÿ to question 22 on page 37

Machine Translated by Google

Question 10 - Answer B is wrong

6 colors are too many.

ÿ to question 10 on page 20

Question 27 - Answer A is correct

The sign says:

Playing ball in the courtyard is permitted, except from 1.00 pm to 4.00 pm and on Sundays.

which means that at 12 you can play ball in the yard, as long as it is NOT Sunday. ÿ to question 30 on page 50

Question 39 - Answer D is correct

Totti pays more for each penalty saved by Buffon than he receives for each penalty scored; since he receives 120 euros from Buffon, the number of penalties scored is greater than those saved.
Totti scored more than 6 penalties and Buffon saved less than 6!
If Buffon saved 5 penalties, Totti would earn

(7 ÿ 40 ÿ 5 ÿ 50) euros = 30 euros

The penalties saved are not only 4; in fact

(8 ÿ 40 ÿ 4 ÿ 50) euros = 120 euros

ÿ to question 13 on page 25

Question 8 - Answer D is wrong

The stars counted are too many.

ÿ to question 8 on page 14

Question 28 - Answer B is wrong

The book is thinner! Look for a method, starting from particular cases (if the book had nine pages, if it had 99…), that allows you to identify the thickness of the book.

ÿ to question 28 on page 44

Question 5 - Answer A is wrong

The law that generates the number of squares has been understood, but the law that forms the numbers written on their faces has not been identified.

ÿ to question 5 on page 11

Machine Translated by Google

Question 13

Which of the following statements is definitely false?

A. Whoever breathes is alive; Piero does not breathe, therefore Piero is dead ÿ page 13 B. No Parisian is Italian; all Parisians speak French; but it is not true that no Italian speaks French C. A square is always a rhombus D. What is written in A. is false ÿ page 31 ÿ page 64 ÿ page 58

E. Every professor has a register; Mario has no register, therefore Mario is not a professor. ÿ page 66

Question 3 - Answer D is wrong

If the solution does not contain sodium it must have potassium and therefore also iodine.

ÿ to question 3 on page 9

Question 2 - Answer E is wrong

Count better, there are fewer triangles.

ÿ to question 2 on page 8

Question 12 - Answer A is wrong

Let's rewrite the hypotheses of the question:

married people > # singles

males > # females

and we add the response condition: • # couples > # single women

If we find at least one particular situation, compatible with the initial hypotheses of the question, which also satisfies the requirements of the answer, then the statement is certainly NOT false and the answer is wrong.

Suppose there are 11 single males and no single females. Married people must be more than 11: let's choose, for example, 6 couples, that is, 12 married people.
We deduce that it is possible that there are more couples than single women.

ÿ to question 12 on page 23

Question 19 - Answer B is wrong

Ruggiero could be behind door 2, but we are not sure.

ÿ to question 19 on page 34

Machine Translated by Google

Question 14

It is read on the bus

Passengers are required to pay an additional fee for each piece of baggage that exceeds at least one of the following dimensions: 50 cm × 30 cm × 25 cm.

Based on this rule it can be deduced that:

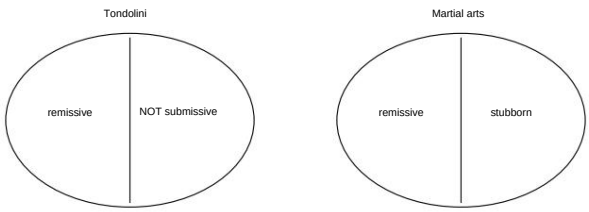
A. you do not have to pay an additional ticket for an object that exceeds 50 cm in length, 30 cm in height and 25 cm in thickness ÿ page 42 B. you do not have to pay an additional ticket for a stick that is 90 cm long and has a diameter of 2 cm if held horizontally ÿ page 15 C. you do not have to pay an additional ticket for a stick that is 90 cm long ÿ page 77 D. you do not have to pay an additional ticket for an object that exceeds 50 cm and with a diameter of 2 cm

in length or 30 cm in height or 25 cm in thickness ÿ page 22

And you don't have to pay an additional ticket for a ball of the diameter ÿ page of 20 cm

49

Question 29 - Answer D is wrong



It could be true; to say that there are no Tondolinis who are also Marziolans is equivalent to saying that the set of Tondolinis and that of Marziolans are disjoint, and this is not incompatible with the hypotheses of the question.

ÿ to question 29 on page 46

Question 5 - Answer D is wrong

The chosen square is equal to the second in the ordered set of squares, but the third is not equal to the first: so it does not appear that the squares alternate!

ÿ to question 5 on page 11

Machine Translated by Google

Question 15

Two players, Aldo and Bruno, take turns placing their pieces on a 3 × 3 board;
the two players' pieces are the same, let's say all black.
The first player to complete a trio in a horizontal, vertical or diagonal row wins.

How many moves can the first player (Aldo) start the game with?
to be able to guarantee victory regardless of how Bruno plays?

A. 8 ÿ page 49

B. 1 ÿ page 48

C. 0, in the sense that Bruno can always respond appropriately
and secure victory D. 5 ÿ page 12

ÿ page 10

E. 9 ÿ page 60

Question 33 - Answer B is wrong

This statement is compatible with both the conjecture and its denial (if they are in
finite number, there is at least one): therefore it does not allow to refute or accept the conjecture.

ÿ to question 33 on page 53

Question 34 - Answer B is wrong

Every well-structured question should have only one correct answer, but this is not it: 4 possibilities
there are too many.

ÿ to question 34 on page 54

Machine Translated by Google

Question 16

Indicate how many different numbers can be obtained from algebraic sums of this type

±1 ± 2 ± 3 ± 4 ± 5

using all the digits from 1 to 5 and varying all the possible choices of the + or ÿ signs (for example: ÿ1 ÿ 2 + 3 + 4 + 5, 1 + 2 ÿ 3 + 4 + 5, …).

A. 20
B. 24
C. 31
D. 16
E. 32

ÿ page 45 ÿ page 23 ÿ page 73 ÿ page 78 ÿ page 73

Question 34 - Answer A is correct

The third digit is half of the fourth, so the fourth is even! It cannot be 4, but it must be greater than 4 in order for the three preceding digits to be smaller (the safe combination digits are increasing).

- The fourth digit is 6. It cannot be 8, because in third position there cannot be a 4
- the third digit is 3
- the second digit, less than 3, can be 2 or 1: zero can only occupy the first position
- the first digit is 1 or 0, if there is 2 in the second position; 0 if there is 1 in the second position

There are three possible combinations and three is the minimum number of attempts for Nonno Peperino:

(0, 2, 3, 6), (1, 2, 3, 6), (0, 1, 3, 6)

ÿ to question 7 on page 13

Question 17 - Answer A is wrong

5 becomes 6 by adding just one more rod to the display.

ÿ to question 17 on page 30

Question 30 - Answer E is wrong

The correct answer is another. Try to write some possible arrangements of the 5 numbers that satisfy the premises; you will see that there are certainly more than six, even if you do not write them all.

ÿ to question 30 on page 50

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Question 30 - Answer A is wrong

This is not the correct answer. Try to write some possible arrangements of the 5 numbers that satisfy the premises; you will see that there are certainly more than three, even if you do not write them all.

ÿ to question 30 on page 50

Question 44 - Answer D is wrong

A third of the first-term parliamentarians are 10 politicians: they could all be unindicted or indicted, but not definitively convicted. ÿ to question 44 on page 71

Question 31 - Answer B is wrong

Rocco is the least rich.

ÿ to question 31 on page 51

Question 27 - Answer D is wrong

From 1pm to 4pm you may NOT play ball in the yard, on any day of the week. So at 2pm playing ball in the yard is prohibited.

ÿ to question 27 on page 43

Question 11 - Answer A is wrong

What relationship does the number 14 have with the other terms of the first sequence?

ÿ to question 11 on page 22

Question 19 - Answer A is wrong

Since one dragon is behind an odd door, the other two cannot be behind an even door.
Ruggiero is therefore behind an equal goal.

ÿ to question 19 on page 34

Question 49 - Answer C is wrong

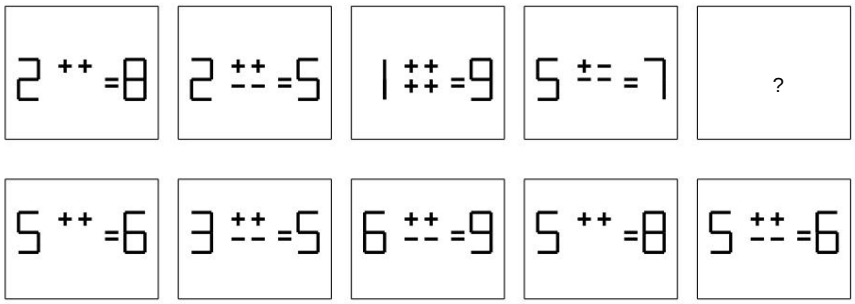
If it is false that Umberto has at least one blond child, then Umberto has no children or has children who are, for example, red-haired.

ÿ to question 49 on page 76

Machine Translated by Google

Question 17

Which of the boxes numbered 1 to 5 replaces the box marked with the dot?
question?



A. 1 ÿ page 28
B. 2 ÿ page 80
C. 3 ÿ page 18
D. 4 ÿ page 41
E. 5 ÿ page 23

Question 20 - Answer D is wrong

The hypothesis

it is not true that at least one male is not of age

allows us to conclude that all males are adults. Therefore there are no married males or single, not of legal age.

ÿ to question 20 on page 35

Question 23 - Answer E is wrong

The first hypothesis of the question

Anyone who listens to rock or blues music is not tone deaf

This is equivalent to saying that someone who is tone deaf doesn't listen to rock music or blues music.

Agenore is not tone deaf and can, if he wants, listen to rock and/or blues music: therefore it is not excluded that you listen to rock.

ÿ to question 23 on page 38

Machine Translated by Google

Question 8 - Answer A is correct

Only two stars satisfy the four conditions.

ÿ to question 2 on page 8

Question 26 - Answer D is correct

Kontakerikonta, although not knowing whether the Incredible numbers are an infinite or finite quantity, formulates a conjecture, a hypothesis about the Incredible numbers, if they are infinite in number.
The student Rikontoijo discovers that the conjecture is false, that is, that the Incredible numbers are infinite, but do not have the property hypothesized by Kontakerikonta: therefore he proves that the Incredible numbers are infinite and none of them has eight distinct prime factors.

ÿ to question 9 on page 15

Question 28 - Answer D is wrong

A bit too many, a figure would have been used for each page and not a number per page!
You probably got the concept of number confused with the concept of digits in a number!

ÿ to question 28 on page 44

Question 11 - Answer E is wrong

What is the relationship between the number 240 and the other numbers in the second

sequence? ÿ question 11 on page 22

Question 9 - Answer A is wrong

Luigina herself says that on Tuesdays, if she takes a bath, then she goes to the market.

ÿ to question 9 on page 15

Question 13 - Answer B is wrong

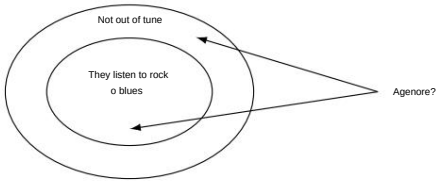
as the statement: no Parisian
is Italian; all Parisians speak French, but it is not true that no
Italian speaks French is true; we are, however,
looking for the reasoning that is certainly false!

ÿ to question 13 on page 25

Machine Translated by Google

Question 23 - Answer A is correct

Let's represent the first hypothesis of the question using Venn diagrams:



The first hypothesis of the question is

that those who listen to rock or blues music are not tone deaf.

This is equivalent to saying that someone who is tone deaf doesn't listen to rock music or blues music.

Agenore is not tone deaf so he can listen to rock or blues music.

Therefore, the proposed answer is certainly incompatible with the premises.

ÿ to question 29 on page 46

Question 38 - Answer A is wrong

Charlemagne is the one who won the most.

ÿ to question 38 on page 63

Question 18 - Answer E is wrong

You have been fooled and, surely, you are in good company!

To be sure that there are two glasses on the table and only two, the statement should have said: exactly two.

Are you sure that there are two glasses, but there could be three, half a dozen… ÿ to

question 18 on page 33

Question 24 - Answer A is wrong

If we find an easy counterexample, we are sure that the statement < is false. For example, if the table is:

(1324) ÿ

1 1	1 3 3 3				
				2 4 3	

ÿ 3 = =

ÿ to question 24 on page 39

Question 51 - Answer D is wrong

If the first player takes 2 pebbles, there are 6 left on the table and nothing prevents the second player from winning by taking them all.

ÿ to question 51 on page 79

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Question 18

The sentence There are two glasses on the table implies that on the table:

A. there are two glasses and a bottle B. there are no bottles C. there are two glasses and two coffee cups D. there is a glass And there are no three glasses

ÿ page 61 ÿ page 43 ÿ page 73 ÿ page 10 ÿ page 32

Question 49 - Answer D is wrong

The denial of:

Umberto has at least one blond son

And

if Umberto has children, then all of them are NOT blond, that is, no one is blond.

The proposed answer suggests instead that there could be blond children and non-blond children. ÿ to question 49 on page 76

Question 19 - Answer C is wrong

Ruggiero could be behind goal 0, but we are not sure about that.

ÿ to question 19 on page 34

Question 7 - Answer E is wrong

The hypothesis

If you do NOT use a gas oven then the meat is either raw inside or burnt on the outside or both

is equivalent to

with electric oven (not gas) ÿroast with meat raw inside or burnt outside or both With the electric oven the meat could also be raw inside or burnt outside without excluding the possibility that it is raw inside and burnt outside.

ÿ to question 7 on page 13

Question 21 - Answer C is wrong

We know that at least one of the seven candidates fears not being among the top three, but we do not know if any of them believe they will be among the top three.

ÿ to question 21 on page 36

Machine Translated by Google

Question 19

The warrior Bradamante finds herself in a room shaped like a regular hexagon, with doors numbered in succession from 0 to 5. Behind each door hides one and only one of the following: three deadly dragons, the beloved paladin Ruggiero, the hippogriff and the mocking wizard Atlante. If an even door hides a dragon, all even rooms hide dragons. Behind door 4, the hippogriff paws the ground, while a dragon's tail peeks out from under the door opposite it.
The magician says: the sum of Ruggiero's door and the two doors adjacent to it is 6.
Bradamante can certainly say that:

A. Ruggiero is behind an odd door B. Ruggiero is behind door 2 C. Ruggiero is behind door 0 D. the magician has not given any new instructions E. Ruggiero is behind door 0 or behind door 3

ÿ page 29 ÿ page 25 ÿ page 33 ÿ page 53 ÿ page 19

Question 39 - Answer B is wrong

If Buffon had saved 5 penalties, Totti would have earned

(7 ÿ 40 ÿ 5 ÿ 50) euros = 30 euros

ÿ to question 39 on page 64

Question 42 - Answer E is wrong

The first hypothesis

if there are few of you, you eat well,

it doesn't say how you eat if there are many of you.
The second hypothesis

if there are many of you you

spend little it does not provide indications on how to eat.
Mr. Aquilotto is wrong to deduce that if there are NOT few (i.e. many) then one does NOT eat well (i.e. badly).
Let's convince him, with the laws of mathematical logic: space =

there are few of us and = we eat well.

we know that

ÿ NOT equivalent to (NOT) ÿ (NOT)

So:

You are NOT few ÿ/ You do NOT eat well.

ÿ to question 42 on page 68

Machine Translated by Google

Question 20

It is known about a family that:

- at least one male is not celibate
- all graduates are single
- it is not true that at least one male is not of age.

Only one of the following propositions can be deduced from the premises. Which one?

A. No adult is unmarried B. All single people are graduates C. At least one adult is married D. At least one single person is not an adult E. At least one adult is not married

ÿ page 58 ÿ page 63 ÿ page 19 ÿ page 30 ÿ page 48

Question 25 - Answer A is wrong

From the hypotheses of the question we know that the defendant defends himself vigorously declaring his innocence, but we do not know if he is telling the truth or if he is lying! ÿ to question 25 on page 40

Question 26 - Answer B is wrong

The answer talks about a set of numbers not taken into consideration by the professor. ÿ to question 26 on page 42

Question 9 - Answer C is wrong

From the hypotheses we know that:

yesterday (Wednesday) Luigina didn't bathe and went to the market.

Luigina says what she did yesterday, Wednesday, but she doesn't say what she usually does on Wednesdays, nor does she tell us whether or not, when she bathes on Wednesdays, she then goes to the market.

ÿ to question 9 on page 15

Question 3 - Answer A is wrong

If it contains potassium it must also contain iodine.

ÿ to question 3 on page 9

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Question 21

In the primary elections to choose the candidate for Prime Minister of Burgundy, each of the seven candidates is sure to be among the three most voted. To deny this sentence is to affirm that:

A. there is at least one candidate who fears being among the three least voted ÿ page 65 B. there is at least one candidate who is not sure of being among the top three most voted C. some of the seven ÿ page 46
candidates are sure of being able to rank among the top three most voted
the three most voted ÿ page 33
D. each candidate is sure that he will not be able to rank among the top three
voted for ÿ page 18 And
each of the seven candidates fears being among the three least voted ÿ page 46

Question 2 - Answer D is wrong

It counts better.

ÿ to question 2 on page 8

Question 3 - Answer B is wrong

The solution must have either sodium or potassium or both.
So at least one of the two substances must be there.

ÿ to question 3 on page 9

Question 40 - Answer D is correct

If Bianchi is not left-handed, Lieutenant Piccione can deduce that Bianchi could not have stabbed the victim, without excluding, however, that he could also be the murderer.
We remind you that:

ÿ is equivalent to (NOT) ÿ (NOT)

ÿ to question 18 on page 33

Machine Translated by Google

Question 22

A survey conducted among all 1000 inhabitants of the town of Sparagna over the age of 55 shows that 40% of them own a refrigerator, 25% own a television, and 15% own both.

So, in the town of Sparagna:

A. less than 40% of the population under 55 owns a refrigerator B. less than 45% of the ÿ page 62

population over 55 owns a television or refrigerator C. 50% of the population over 55 owns neither a ÿ page 23

television nor a refrigerator ÿ page 79

D. under 55 years of age more than 25% of the population owns a television ÿ page 78

E. more than 50% of the population over 55 years old has a television or refrigerator ÿ page 39

Question 3 - Answer E is correct

The possible substances contained in the solution are:

- sodium
- sodium, jodium
- sodium, iron
- sodium, ferro, jodium
- sodium, potassium, iron, iodine
- potassium, iron and iodine
- sodium, potassium and iodine

The solution can therefore only contain sodium.

ÿ to question 23 on page 38

Question 11 - Answer B is wrong

What relationships are there between the numbers 14 and 240 with the first or second sequence of numbers respectively?

ÿ to question 11 on page 22

Machine Translated by Google

Question 23

Given that:

- those who listen to rock or blues music are not tone deaf
- Agenore is not out of tune
- those who listen to the blues don't win the lottery

Which of the following conclusions CANNOT be drawn from the previous premises?

A. It is impossible for Agenore to listen to blues B. A tone deaf person does not listen to rock

C. It is possible that Agenore will not win the lottery

D. Lottery winners don't listen to the blues

E. It is not excluded that Agenore listens to rock

ÿ page 32
ÿ page 68
ÿ page 19
ÿ page 22
ÿ page 30

Question 25 - Answer B is wrong

Let's rephrase the hypothesis. Say that:
The commissioner does not believe that the accused of murder did not go to the victim's home in the time interval in which she lost her life

it is equivalent to saying that:
the commissioner believes that the defendant went to the house in the time interval in where he lost his life.

ÿ to question 25 on page 40

Question 12 - Answer D is correct

Let's rewrite the hypotheses of the question:

married people > # singlesÿ # singles < 2(# couples)

males > # females

and add the response condition:

- # single > 2(# coppie)

Since the number of married people is double the number of couples, the first hypothesis
This is equivalent to saying that the number of singles is less than (not more than!) double the number of couples.
The statement is therefore certainly FALSE.

ÿ to question 10 on page 20

Machine Translated by Google

Question 24

I consider a square table made up of 4 different numbers and arranged in 2 rows each composed of 2 numbers:

( )

Let us be:

- $_1$ the smallest number in the first row
- $_2$ the smallest number in the second row
- the greatest among $_1$ ed $_2$
- $_1$ the largest number in the first column
- $_2$ the largest number in the second column
- the lesser of $_{1 \text{ and } 2}$

So we can conclude that:

A. <

B. =

C. >

D. ÿ

E. ÿ

ÿ page 32
ÿ page 40
ÿ page 50
ÿ page 47
ÿ page 9

Question 22 - Answer E is wrong

The number of citizens of Sparagna over 55 years old with a television or refrigerator, taking into account that 15% own both, is:

400 + 250 ÿ 150 = 500 ÿ 50%

The answer is wrong because it provides a percentage strictly greater than 50%! (more than 50%) and not 50%.

ÿ to question 22 on page 37

Question 28 - Answer A is wrong

Try to count the pages by dividing them into blocks, based on the digits of each number: pages and 1 in 9; and 10 a...

ÿ to question 28 on page 44

Machine Translated by Google

Question 25

The lady she was strangled in her living room. The commissioner does not believe that the accused of murder (who vigorously defends himself by declaring his innocence) did not go to the victim's house during the time in which she lost his life. It can be deduced that:

A. the defendant is certainly not guilty ÿ page 35

B. the commissioner he is convinced that the accused did not go at the victim's home during the time period in which the victim he lost his life ÿ page 38

C. for the commissioner the defendant is certainly not guilty ÿ page 70

D. the defendant is certainly guilty ÿ page 78

And the commissioner does not exclude that the defendant strangled the lady ÿ page 66

Question 24 - Answer B is wrong

If we find an easy counterexample, we are sure that the statement = is false. If the table

And:

(3 2 1 4) ÿ

1	2				
1 1	2 2 3			2 4 3	

ÿ 2 = < = 3

ÿ to question 24 on page 39

Question 40 - Answer C is wrong

Let's rewrite the first hypothesis:
if he stabbed the victim, then he is left-handed
Attention, the error is classic:

ÿ is NOT equivalent to ÿ

The stabber must be left-handed, but this does not mean that all left-handed people (for example the Mr. Rossi) are stabbers, much less murderers.

ÿ to question 40 on page 65

Question 37 - Answer B is wrong

Consistent with the assumptions of the question, if the budget was not cut and all taxes were prices then increased and had to remain stable.

ÿ to question 37 on page 59

Machine Translated by Google

Question 17 - Answer D is correct

If the symbols + + - - indicate how many rods must be turned on or off to display the number indicated beyond the = sign, with just two more rods the 5 becomes 8! ÿ on page 81

Question 8 - Answer B is wrong

Reread the questions and count them again.

ÿ to question 8 on page 14

Question 42 - Answer D is wrong

The second hypothesis of the question

if there are many of you, you spend little.

It does not mean that you spend little only if there are many of you.

Mr. Aquilotto cannot deduce that, in order to spend little (individual spending), there must be many of us: he can hope that the price is low even if he goes to dinner with a few friends.

Mr. Aquilotto can be convinced of the groundlessness of his reasoning with the help of mathematical logic by posing

= there are many of us and = we spend little

and reminding him that

ÿ is NOT equivalent to ÿ

ÿ to question 42 on page 68

Question 38 - Answer D is correct

If with GC, N, CM, AM, … we indicate the patrimony of Julius Caesar, Napoleon, Charlemagne, Alexander the Great, … we can write the following inequalities:

ÿ > > >

Comparing the relations we deduce that

> > ÿ >

Augusto is the one who won the least!

ÿ to question 18 on page 33

Question 46 - Answer E is wrong

Cyrillic numbers are much less numerous!

ÿ to question 46 on page 73

Machine Translated by Google

Question 26

The great number theorist Kontakerikonta discovered the Incredible Numbers; he does not yet know whether they are finite in number, but he has made the following conjecture:

- if they are infinite, at least one of them must have 8 distinct prime factors

His student Rikontoijo, carefully studying these numbers, proves that Kontakerikonta's conjecture is false.

So Rikontoijo has proven that:

A. if the Incredible numbers are a finite quantity, none of them have 8 distinct prime factors ÿ page 66

B. if the Incredible numbers are a finite quantity, they all have 8 distinct prime factors ÿ page 35

C. The Incredible Numbers are infinite ÿ page 74

D. The Incredible Numbers are infinite and none of them have 8 factors distinct primes ÿ page 31

And. The Incredible numbers are infinite and all have 8 prime factors distinct ÿ page 47

Question 14 - Answer A is wrong

The dimensions of the object indicated in answer A exceed the maximum dimensions indicated in the regulation. Therefore, an additional ticket must be paid for an object with dimensions exceeding 50 cm in length, 30 cm in height and 25 cm in thickness.

ÿ to question 14 on page 26

Question 27 - Answer E is wrong

On Sundays, we do NOT play ball in the yard, neither at 2:00 PM nor at any other time! ÿ to question 27 on page 43

Question 43 - Answer A is correct

The Grand Chamberlain advises the King to oppose the marriage of the Princess to the Toad if the young people intend to marry in the Basilica of Superga.
The King, not opposing the celebration of marriage in any form, leaves the young people the right to freely decide the place of marriage. ÿ to question 40 on page 65

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Question 27

In the entrance hall of an apartment building there is a sign with the following notice:

Playing football in the courtyard is permitted, except from
1.00 pm to 4.00 pm and on Sundays.

It can be deduced that in that condominium:

A. It is not forbidden to play football in the courtyard at 12.00, as long as it's not Sunday ÿ page 24

B. It is not forbidden to play football in the courtyard on Sundays from 4pm onwards ÿ page 17

C. on days other than Sunday it is forbidden to play football in the courtyard before 1.00 pm and after 4.00 pm ÿ page 8

D. It is not forbidden to play football in the courtyard at 2.00 pm, as long as it is not Sunday ÿ page 29

And it is not forbidden to play football in the courtyard at 2.00 pm, as long as it is Sunday ÿ page 42

Question 35 - Answer D is wrong

The figure you are looking for should not be a hexagon.

ÿ to question 35 on page 55

Question 18 - Answer B is wrong

Who assures us that there are NO bottles on the table?

ÿ to question 18 on page 33

Question 51 - Answer C is wrong

If the first player takes 1 pebble, the second player wins the game by taking the 7 pebbles remaining on the table.

ÿ to question 51 on page 79

Question 43 - Answer E is wrong

The King does not want to oppose the celebration of marriage with any formality; young people can decide to marry or not to marry, to marry in the Basilica of Superga or elsewhere. ÿ to question 43 on page 69

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Question 28

A total of 3301 digits were used to number the pages of a book.
The pages of the book are:

A. between 1500 and 2000

B. between 2000 and 3000

C. less than 1000

D. more than 3000

E. between 1000 and 1500

ÿ page 39 ÿ page 24 ÿ page 77 ÿ page 31 ÿ page 75

Question 50 - Answer A is wrong

The sequence of numbers is increasing!

ÿ to question 50 on page 77

Question 10 - Answer D is wrong

5 colors are too many.

ÿ to question 10 on page 20

Question 47 - Answer D is wrong

It is true that the number of days depends on the size of the pond, but knowing that with one archaebacterium it takes 30 days and that the pond is always the same, the time with two archaebacteria can be determined without knowing the size of the pond.

ÿ to question 47 on page 74

Question 35 - Answer E is wrong

The missing figure is a square, but it's not the right one!

ÿ to question 35 on page 55

Question 2 - Answer C is wrong

There are some triangles missing.

ÿ to question 2 on page 8

Question 1 - Answer B is wrong

It's not possible: Bruno doesn't know how to drive and so he needs a van driver.

ÿ to question 1 on page 7

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Question 7 - Answer B is wrong

The hypothesis

If you do NOT use a gas oven then the meat is
either raw inside or burnt on the outside or both

equivalent

to a roast well cooked inside and not burnt on the surface ÿgas oven
Indeed:

ÿ is equivalent to (NOT) ÿ (NOT)

that's it

= ÿ then (NOT) = (NOT) ÿ (NOT)

So if q is: (raw

roast inside) or (burnt on the surface) or (both)

then (NONq) becomes:

(roast well done inside) and (not burnt on the surface).

To be sure that the roast has been cooked in a gas oven the meat must have
the inside well cooked and the outside not burnt!

ÿ to question 7 on page 13

Question 16 - Answer A is wrong

It is not the exact number: reflect starting from particular cases.

ÿ to question 16 on page 28

Question 45 - Answer B is wrong

Note that the first number of each pair belongs to an increasing sequence, while the second to a decreasing one.

ÿ to question 45 on page 72

Question 38 - Answer C is wrong

Julius Caesar won less than Napoleon.

ÿ to question 38 on page 63

Question 31 - Answer D is wrong

Silvio is the richest.

ÿ to question 31 on page 51

Question 3 - Answer C is wrong

If it contains potassium it must also contain iodine.

ÿ to question 3 on page 9

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Question 29

A Marziolan, observing that:

- half of all Tondolini are submissive
- half of all Marziolans are stubborn
- half of all Marziolans are submissive

and keeping in mind that one cannot be both submissive and stubborn at the same time, he deduces that one and only one of the following statements CANNOT be true. Which one?

A. Half of all Tondolini are stubborn ÿ page 52 ÿ page

B. All Tondolini are Marziolani 66 C. All Marziolans

are Tondolini and no Tondolino is stubborn ÿ page 47 D. There are no Tondolini who are also Marziolans ÿ page 26 ÿ page 61

E. Tondolini and Marziolani are the same group of people

Question 21 - Answer B is correct

Let's rewrite the hypothesis.

All seven candidates are sure to rank among the top three

The negation of the statement becomes:

There is at least one candidate who is NOT certain to be among the top three voters

ÿ to question 44 on page 71

Question 21 - Answer E is wrong

There are two errors made in formulating the answer. We know that

1. at least one of the seven candidates fears NOT being among the three most voted, but we are not sure that all seven fear NOT being in the top three in the standings
2. Not being in the top three in the ranking means placing yourself among the four least voted

ÿ to question 21 on page 36

Question 6 - Answer D is wrong

Romulus and Remus would both be liars: the hypotheses of the question ensure that only one of the two is a liar!

ÿ to question 6 on page 12

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Question 29 - Answer C is correct

The statement is certainly false: to say that all Marziolans are Tondolini is equivalent to saying that the set of Marziolans is a subset of that of Tondolini and therefore the stubborn Marziolans are a subset of the NON-submissive Tondolini; therefore there exist non-submissive Tondolini who are stubborn.

ÿ to question 12 on page 23

Question 1 - Answer E is wrong

Ciro only works with Aldo!

ÿ to question 1 on page 7

Question 7 - Answer D is wrong

The hypothesis states that the cook Giovanni found that
without a gas oven ÿthe roast is raw internally
or burned externally or both

In mathematical logic

ÿ is NOT equivalent to ÿ

So a roast that is raw inside does not necessarily have been cooked without a gas oven.
A gas oven is necessary to cook a roast well, but it is not sufficient. Other conditions may be required: for example, that the cook puts the oven at the right temperature, leaves the roast in the oven for the necessary time, etc.

ÿ to question 7 on page 13

Question 26 - Answer E is wrong

Kontakerikonta formulates a conjecture about the Incredible infinite numbers in number
at least one of them must have 8 distinct prime factors.
If the conjecture is false then
the set of Incredibles is an infinite quantity and there is no
Incredible number with 8 distinct prime factors

ÿ to question 26 on page 42

Question 24 - Answer D is wrong

This counter-example assures us that it is not true that ÿ

(3214) ÿ

1 1 1 2 2 3					
				2 4 3	

ÿ 2 = < = 3

ÿ to question 24 on page 39

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Question 4 - Answer E is wrong

The answer is wrong. The second numbers of each pair appear to be arranged in ascending order…

ÿ to question 4 on page 10

Question 33 - Answer E is wrong

To say that there exists a positive integer such that, for every number (prime or non-prime) with > the number + 2 is not prime is equivalent to saying that prime numbers are finite!

ÿ to question 33 on page 53

Question 48 - Answer E is wrong

It is a thought of common language, not logically deducible from the hypotheses of the question.
It could be a statement made between friends.

ÿ to question 48 on page 75

Question 20 - Answer E is wrong

Let's rewrite the hypotheses of the question:

- at least one male is married
- all graduates are not married
- all males are adults

We can deduce that:

all males are of legal age and at least one male is married.

At least one could mean everyone, so we have no way of saying whether there are celibate males.

ÿ to question 20 on page 35

Question 34 - Answer E is wrong

The right answer to a question is always unique, but in this case it is another. Grandpa Peperino opens the safe with a maximum of two attempts only if he is lucky. Remember that Grandpa Peperino can also use zero.

ÿ to question 34 on page 54

Question 15 - Answer B is wrong

The answer is wrong. You should try to draw a chessboard and do some simulations!
Play to convince yourself!

ÿ to question 15 on page 27

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Question 30 - Answer B is wrong

The correct answer is another one. Don't they seem like a lot to you? There are 24 permutations of four objects in any order, but if you put the two even numbers only in the even places, and the odd numbers only in the odd places, you don't have that many permutations to do.

ÿ to question 30 on page 50

Question 14 - Answer E is correct

The diameter of the ball does not exceed any of the three dimensions indicated in the standard. Furthermore, given its spherical shape, the ball can be contained in a box measuring 20 cm x 20 cm x 20 cm, or having dimensions that do NOT exceed the three maximum dimensions indicated in the standard. Therefore, the passenger who intends to transport a ball with a diameter of 20 cm does not need to purchase an additional ticket.

ÿ to question 37 on page 59

Question 39 - Answer A is wrong

If Buffon had saved 9 penalties, Totti would be at a loss!

ÿ to question 39 on page 64

Question 45 - Answer E is wrong

Neither number is correct. This is not the correct law. The second number of the pair should be between 19 and 13.

ÿ to question 45 on page 72

Question 38 - Answer B is wrong

There are those who have won less than Alexander the Great.

ÿ to question 38 on page 63

Question 15 - Answer A is wrong

In each question there is only one correct answer and it is not this one. There are 8 possibilities but they are not the right number. Play to convince yourself!

ÿ to question 15 on page 27

Question 44 - Answer C is wrong

40 parliamentarians were investigated and definitively convicted, the other 80 were not investigated or, if investigated, were not definitively convicted. Choosing three parliamentarians at random, we could just come across three politicians belonging to the group of 80.

ÿ to question 44 on page 71

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Question 30

Wanting to arrange the numbers

28 29 36 43 55

so that the odd ones occupy an odd position and the even ones occupy an even position equal, how many different ways can this be done?

A. 3 ÿ page 29
B. 24 ÿ page 49
C. 12 ÿ page 80
D. 5 ÿ page 50
E. 6 ÿ page 28

Question 24 - Answer C is wrong

If we find an easy counterexample, we are sure that the statement > is false. Consider the table:

(1324) ÿ

1	2				
1 1	3 3 3			2 4 3	

ÿ 3 = =

ÿ to question 24 on page 39

Question 43 - Answer B is wrong

The King does not listen to the advice of the Grand Chamberlain and the princess and the toad can marry where and when they want.

ÿ to question 43 on page 69

Question 36 - Answer E is wrong

The correct answer is another: with no 20 euro piece would he not have 195 euros in his cash register.

ÿ to question 36 on page 56

Question 30 - Answer D is wrong

Only one of the answers is correct and not this one! Try to write some possible arrangements of the 5 numbers that satisfy the premises; you will see that there are certainly more than five, even if they are not the write them all.

ÿ to question 30 on page 50

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Question 31

There are five people with different financial situations.
Oronzo is richer than Rocco, whose wealth is more modest than that of Silvio, and the latter in turn is wealthier than Piero. Quirino is less well-off than Piero, but more comfortable than Oronzo.

Who is the third richest?

A. Piero
B. Rocco
C. Oronzo
D. Silvio
E. Quirino

ÿ page 66 ÿ page 29 ÿ page 54 ÿ page 45 ÿ page 21

Question 10 - Answer A is wrong

4 colors are too many.

ÿ to question 10 on page 20

Question 45 - Answer C is wrong

Only one of the two numbers in the pair is correct: the second.

ÿ to question 45 on page 72

Question 36 - Answer B is wrong

That's not the right answer. You can't have 4 pieces of 20 euros; what pieces should the remaining 115 euros be made of?

ÿ to question 36 on page 56

Question 42 - Answer B is wrong

The hypothesis

if there are few of you, you eat

well does not say you eat well only if there are few of you: Mr. Aquilotto cannot, therefore, deduce that to eat well it is necessary to go to the Oca Giuliva with a few people. He can hope that even a cheerful group can eat well in company.
From the point of view of mathematical logic if

=you spend little and = you eat well

so long as

ÿ is NOT equivalent to ÿ

Mr. Aquilotto has no information to deduce that if you eat well then you are few.

ÿ to question 42 on page 68

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Question 32

Examine the following chain of propositions

Only engineers are reliable people
Some reliable people are involved in politics

it can be deduced that:

A. nothing can be deduced ÿ page 23 B. no engineer deals with politics ÿ page 74 C. anyone who deals with politics is not a trustworthy person ÿ page 7 D. some engineers deal with politics ÿ page 61 E. all engineers deal with politics ÿ page 16

Question 29 - Answer A is wrong

It could be true; it is possible that half of the Tondolini about whom the text says nothing, but who are not submissive, are stubborn.

ÿ to question 29 on page 46

Question 7 - Answer C is correct

Let's interpret the hypothesis of the question

If you do NOT use a gas oven then the meat is
either raw inside or burnt on the outside or both

with the language of mathematical logic.
Knowing that

ÿ is equivalent to (NOT) ÿ (NOT)

and what if

= ÿ then (NOT) = (NOT) ÿ (NOT)

The hypothesis

NOT (gas oven) ÿ(raw meat inside) or (burnt outside)
or (both)

is equivalent to
NOT [(raw meat inside) or (burnt outside) or (both)] ÿNOT (NOT gas oven) which we can rewrite (well done
meat inside) and (not
burnt outside) ÿgas oven

ÿ to question 26 on page 42

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Question 33

A famous conjecture states that

there are infinitely many prime numbers such that **+ 2** is a prime number.

To refute this statement is equivalent to proving that

A. for every positive integer and for every prime number with > the number + 2 is not prime ÿ page 68

B. there exists a positive integer and a prime number with > such that the number + 2 is not prime ÿ page 27

C. for every positive integer there exists a prime number with > such that the number + 2 is not prime ÿ page 7

D. there exists a positive integer such that, whatever the number prime with > the number + 2 is not prime ÿ page 71

And. there exists a positive integer such that, for every number (first and not first) with > the number + 2 is not prime ÿ page 48

Question 19 - Answer D is correct

If a hippogriff is pawing the ground behind an even door, then dragons must be behind the odd doors (1, 3, 5).
On the other hand, a dragon's tail is peeking out from behind door 1.
Ruggiero, therefore, must be behind an even goal other than the 4th, behind which he is pawing.
the hippogriff. Ruggiero is therefore behind gate 0 or 2.
The mocking magician Atlas does not provide any further information, because the only three consecutive doors
with the sum of the three numbers 6 are the gates with numbers (5, 0, 1) or (1, 2, 3)
Atlante confirms, therefore, the information already in Bradamante's possession: Ruggiero is behind
to the only even port of each triplet: port 0 or port 2.

ÿ to question 1 on page 7

Question 43 - Answer D is wrong

The data in the question authorise us to conclude that one of the proposed interpretations is true and that
falsity of four others (including this one).

ÿ to question 43 on page 69

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Question 34

Grandpa Peperino no longer remembers the combination to his electronic chest.
He only remembers that:

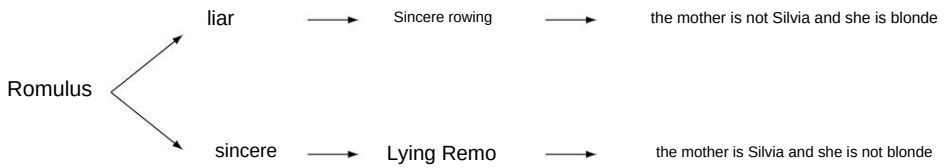
- is made up of four distinct digits between 0 and 9
- the 4 doesn't appear
- the third digit is half of the fourth
- the numbers increase from first to last

How many possible combinations are there for Grandpa Peperino's chest?

A. 3 ÿ page 28
B. 4 ÿ page 27
C. 5 ÿ page 18
D. 6 ÿ page 60
E. 2 ÿ page 48

NOTE We commonly speak of combinations of a safe; in reality they are provisions, because
It is important to remember not only the numbers, but also their order.

Question 6 - Answer A is correct



So if the mother is blonde, her name is not Silvia (and if her name is Silvia, she is not blonde).

ÿ to question 17 on page 30

Question 31 - Answer C is wrong

Oronzo is not the fourth in order of wealth

ÿ to question 31 on page 51

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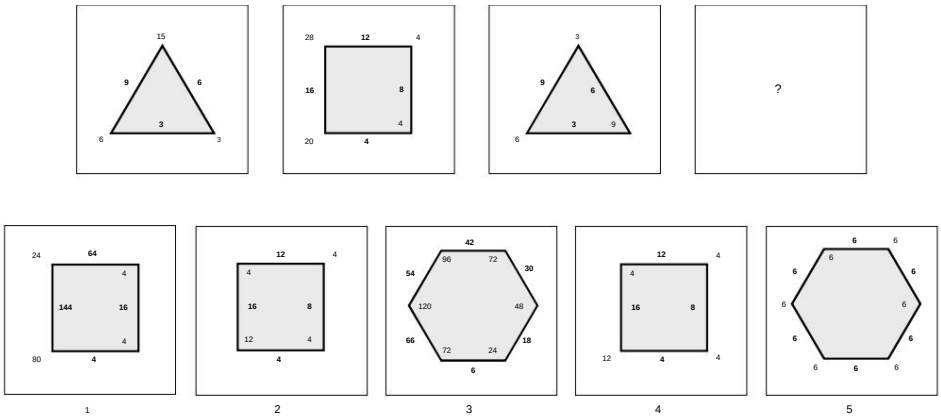
Cheese No. 4

Question 35

Which figure completes the sequence?

Which of the figures numbered from 1 to 5 correctly replaces the box containing the question mark?



A. Figure 2 ÿ page 67

B. B. Figure 3 2 C. ÿ page 8

C. Figure 4 3 4 D. ÿ page 7

D. Figure 5 5 E. Figure 1 ÿ page 43

AND. ÿ page 44

Question 51 - Answer E is wrong

If the first player takes 3 pebbles, the second player can take 5 and wins the game.

ÿ to question 51 on page 79

Question 51 - Answer A is wrong

Play is believe!

ÿ to question 51 on page 79

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Question 36

Franco the tobacconist has 195 euros in his cash register and no coins.

A customer asks him if he has 10, 20, 50 or 100 euros to change, but Franco reluctantly answers no to all requests.

How many 20 euro bills does Franco have in his till?

A. 1 ÿ page 16

B. 4 ÿ page 51

C. 3 ÿ page 62

D. 2 ÿ page 63

E. 0 ÿ page 50

Question 47 - Answer C is correct

Let's try to understand the growth law of archaebacteria.

. batt.	g. 1	g. 2	g. 3	g. 4	… g. 29 g. 30		
1	1	2	4	8 …		2^{28} pond full	
1	2^{0} = 1 2	= 2 2	= 4 2	= 8 …		2^{28}	2^{29}

Let's repeat the calculation with two bacteria, knowing that the full pond must contain 2^{29} archeobat skin

. batt.	g. 1	g. 2	g. 3	g. 4	… g. 28		g. 29
2	2	4	8	16 …		2^{28} pond full	
2	2^{1} = 2 2	= 4 2	= 8 2	= 16 …		2^{28}	2^{29}

We can observe that, starting with two archaebacteria on the first day, the number of bacteria on any given day is the same as there would be with only one initial archaebacterium on day next. Then the pond fills up after 29 days.

ÿ to question 48 on page 75

Question 48 - Answer D is correct

The hypothesis

We all like coffee, except Paola, who doesn't like it.

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he says that

Among all of Antonio's friends, only Paola doesn't like coffee

If this hypothesis is false then:

1.) Paola likes coffee as much as any other friend of Antonio, other than Paola

or

2.) Paola likes coffee and there is at least one other friend of Antonio, other than Paola, who doesn't like it I like coffee

or

3.) Paola doesn't like coffee and there is at least one other friend of Antonio, other than Paola, who I don't like coffee

In the first two cases it is clear that:

Paola likes coffee

while the third hypothesis says that:

there is at least one other friend, besides Paola, who doesn't like coffee

The correct answer is:

either Paola likes coffee, or there is someone among her friends, besides Paola, who doesn't like coffee.

ÿ to question 49 on page 76

Question 37 - Answer C is wrong

According to the assumptions of the question, if the budget was not cut and all taxes were not increased, prices should not remain stable. Prices would remain stable if and only if taxes were increased.

ÿ to question 37 on page 59

Question 8 - Answer E is wrong

Recount better in light of the questions asked.

ÿ to question 8 on page 14

Question 45 - Answer D is correct

The first number of each pair seems to be obtainable by adding a power of 2 to the previous one, in an increasing manner:

5 = 3 + 2, 9 = 5 + 4, ? = 9 + 8, 33 = ? + 16

The number we are looking for is therefore (9+8)=17.

The second number belongs to a decreasing sequence, with the same rule with which the first numbers increase:

27 = 43 ÿ 16, 19 = 27 ÿ 8, ? = 19 ÿ 4, 13 = ? ÿ 2

The number we are looking for is therefore (19-4)=15.

ÿ to question 46 on page 73

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Question 20 - Answer A is wrong

The statement to be evaluated:

no adult is unmarried.

is equivalent to:

all adults are married.

From the hypotheses of the question we know that

- at least one male is married;
- graduates, if they exist, are not married;
- all males are adults.

So at least one male in the family is married, but we cannot be certain that all males are married.

ÿ to question 20 on page 35

Question 39 - Answer C is wrong

If Buffon saved 6 penalties, Totti would have lost 60 euros: in fact, Totti pays Buffon 10 euros more for each penalty saved than he receives from Buffon for each penalty scored! ÿ question

39 on page 64

Question 13 - Answer D is wrong

It is true that the statement is false

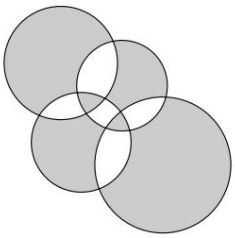
Whoever breathes is alive; Piero is not breathing, therefore

Piero is dead because Piero could be alive and holding his breath. The question asks you to identify the statement that is definitely false and this one is definitely true!

ÿ to question 13 on page 25

Question 10 - Answer C is correct

Let's proceed empirically: let's start from any area and color it with color 1. We insert a new color if and only if we are forced!



ÿ to question 43 on page 69

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Question 37

Matlandia's economy minister says:

> If the budget is not cut, then next year 2006 prices will remain stable if and only if we increase all taxes.

Assuming the absolute truth of this statement and relying solely on it, what could have happened to Matlandia in 2006?

A. The budget was not cut; all taxes were raised and prices remained stable. ÿ page 72

B. The budget was not cut; all taxes were raised and prices rose. ÿ page 40

C. The budget was not cut; taxes were not raised and prices remained stable. ÿ page 57

D. The budget was not cut; taxes were only increased on the salaries of government employees and prices remained stable ÿ page 9

E. The budget was not cut, and prices rose anyway ÿ page 64

Question 8 - Answer C is wrong

Reread the questions carefully and count them again.

ÿ to question 8 on page 14

Question 41 - Answer C is wrong

Let's reformulate the hypothesis.

It is not true that every MP is an honest and competent person, it is equivalent to saying that there is at least one MP (we do not know how many) who is not honest or is not competent or is neither honest nor competent. In Arraffa there may be honest and competent MPs, but we are not sure of it.

ÿ to question 41 on page 67

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Question 49 - Answer B is correct

Umberto has at least one blond son

allows us to deduce that:

- Umberto has at least one child

- at least one of the children is blond

The negation of the statement is:

Umberto has no children or none of Umberto's children are blond.

ÿ to question 31 on page 51

Question 34 - Answer D is wrong

This is not the correct answer: Nonno Peperino needs fewer attempts, if he thinks correctly.
and remember all the conditions.

ÿ to question 34 on page 54

Question 15 - Answer E is correct

Here is the chessboard:

(1,1)	(1,2)	(1,3)
(2,1)	(2,2)	(2,3)
(3,1)	(3,2)	(3,3)

Note that the game choices have an evident symmetry: it makes no difference whether you place the first pawn in

- (1,1), (1,3), (3,1), (3,3)
- (1,2), (2,1), (2,3), (3,2)

Let us therefore consider just one case.

- Aldo has victory in his pocket, whatever Bruno's move, if he places his first piece in (2,2).
- If Aldo places his first piece in a cell on the outer edge, in a corner (for example in (1,1)) or in the centre (for example in (1,2)), he has the possibility of winning with his second or third pawn, whatever cell Bruno chooses.

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– If Bruno places his first piece so that the two pieces on the board are aligned horizontally, vertically or diagonally: the game is done!

– If Bruno places his first piece so that it is not aligned horizontally, vertically or diagonally with his partner's first piece, Aldo must place his second piece in the only empty cell that is at the intersection of two lines (one horizontal and one vertical) that do NOT yet have any pieces and… he does not win on his second move, but on his third.

Play to convince yourself!

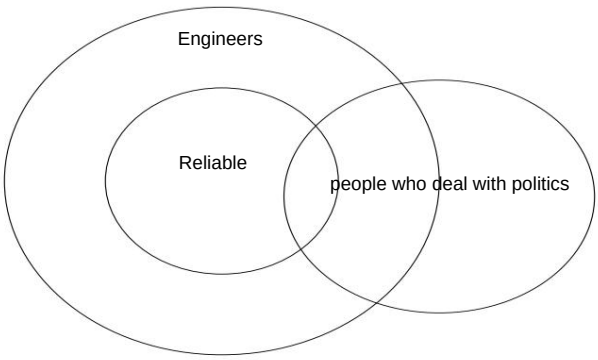
ÿ to question 36 on page 56

Question 5 - Answer C is wrong

The selected square contains too many squares. It would be okay if the square with the question mark was the first one.

ÿ to question 5 on page 11

Question 32 - Answer D is correct



From the hypotheses of the question we deduce that trustworthy people are engineers and that some trustworthy people (therefore some engineers) deal with politics.

ÿ to question 6 on page 12

Question 18 - Answer A is wrong

We have no information to suggest that there was also a bottle on the table…

ÿ to question 18 on page 33

Question 29 - Answer E is wrong

It could be true; nothing prevents the number of Marziolani and Tondolini from being the same and even from being the same group of people, described by different adjectives (submissive, stubborn, etc.), but compatible with each other.

ÿ to question 29 on page 46

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Question 12 - Answer C is wrong

To prove that a certain statement is CERTAINLY NOT false, it is sufficient to find at least one particular case in which that statement is TRUE.

Let's rewrite the hypotheses of the question:

- # married people > # singles
- # males > # females

and we add the response condition: • # husbands > # single women

Let us choose a particular case that satisfies all the hypotheses of the question.
For example, let's take the case of two couples, one single male and zero single females.
There are two husbands, zero single women, so it is true that in Italy there are more husbands than single women.

ÿ to question 12 on page 23

Question 44 - Answer E is correct

Let's summarize the situation in the table

30 parliamentarians in their first term 90 parliamentarians in their first term 80	
40 parl. investigated and convicted	parliamentarians not convicted or sentenced def.

Since only 30 parliamentarians are in their first term and 40 have been definitively convicted, even if all the new parliamentarians had been convicted, necessarily at least 10 parliamentarians, not of first appointment, have been investigated and definitively convicted.

ÿ to question 42 on page 68

Question 36 - Answer C is wrong

This is not the correct answer. If you had three 20 euro bills, what bills would the remaining 135 euros be made up of?

ÿ to question 36 on page 56

Question 22 - Answer A is wrong

The hypotheses of the question do not provide information on the citizens of Sparagna under the age of 55!

ÿ to question 22 on page 37

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Question 38

Playing Risk, Julius Caesar won more than his nephew Augustus, but not than Napoleon.
Alexander the Great conquered less than Charlemagne, but more than Napoleon.
Who won the least?

A. Charlemagne B. ÿ page 32 ÿ page
Alexander the Great C. 49 ÿ page 45 ÿ
Napoleon D. page 41 ÿ page 11
Augustus E.
Julius Caesar

Question 20 - Answer B is wrong

The hypotheses of the question ensure that all graduates are single, but not that all single people are graduates.

ÿ is not equivalent to ÿ

ÿ to question 20 on page 35

Question 36 - Answer D is correct

How many 5 euro, 10 euro… pieces can Franco have in the cash register, without being able to satisfy the customer's requests?

- 5 euro notes: you must have one and only one so you cannot change 10 euros
- 10 euro notes: he can have none or at most one, otherwise he could change 20 euro
- 50 euro notes: you must have one and only one
- 100 euro notes: he must have one, otherwise he wouldn't be able to have 195 euros in the till.

Franco certainly has the following in his pocket:

(1 ÿ 100 + 1 ÿ 50 + 1 ÿ 5) euros = 155 euros

The missing 40 euros can only be the sum of two 20 euro notes.

ÿ to question 34 on page 54

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Question 39

At the end of a training session of the national team, Totti and Buffon make the following bet: Totti will take 12 penalties and Buffon will try to save them. For each penalty saved, Totti will give 50 euros to Buffon while for each penalty scored, Buffon will give 40 euros to Totti. After that, the series of penalties is carried out at the end of which Totti must receive 120 euros from Buffon.

How many penalties has Buffon saved?

A. 9
B. 5
C. 6
D. 4
E. 12

ÿ page 49 ÿ page 34 ÿ page 58 ÿ page 24 ÿ page 77

Question 6 - Answer E is wrong

Romulus would certainly be a liar and the hypotheses of the question do not give us this certainty: one of the two is a liar, but we do not know who!

ÿ to question 6 on page 12

Question 46 - Answer B is correct

Adding any Cyril number to itself, by the first property, I find a Cyril. If I add to the new result again, a new Cyril is generated. I repeat the procedure times, and I find the number:

= + + ÿ + = ÿ

It is a Cyril number for the first property, but it is not a Cyril number for the second property.
The number of Cyrillic numbers is zero (!)

ÿ to question 3 on page 9

Question 13 - Answer C is wrong

The proposed statement is true: a rhombus is a quadrilateral with 4 sides that are congruent to each other and two by two parallel; the square is a particular rhombus with four angles that are all congruent to each other. The question is looking for the statement that is definitely false!

ÿ to question 13 on page 25

Question 37 - Answer E is wrong

The proposed answer, while stating that the budget was not cut, does not provide information on what was decided for taxes! So we cannot deduce whether prices remained stable or not. ÿ to question 37 on page 59

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Question 40

Lieutenant Piccione, during his investigations into a murder, ascertained these two facts:

- if he stabbed the victim, then he is left-handed;
- if he stabbed the victim, then he is the murderer.

Which of these deductions is correct?

A. Commissioner Piccione ascertains that Mr. Bianchi is not left-handed ÿ page 21
and deduces that he is not the murderer

B. The killer stabbed the victim ÿ page 18

C. Commissioner Piccione ascertains that Mr. Rossi is left-handed and
deduces that he is the murderer ÿ page 40

D. Commissioner Piccione ascertains that Mr. Bianchi is not left-handed ÿ page 36
and deduces that he did not stab the victim

E. Commissioner Piccione ascertains that Mr. Rossi is left-handed
and deduces that he stabbed the victim ÿ page 16

Question 1 - Answer D is wrong

If Bruno is paid, then with him there was the driver who cannot be Aldo (unless Aldo works for free). But if Ciro is the driver then there is also Aldo… who must be paid.

ÿ to question 1 on page 7

Question 21 - Answer A is wrong

Denial: Each of
the seven candidates is certain to rank among the top three

And

at least one of the seven candidates believes he will NOT be among the three most voted. Saying that at least one candidate fears not being among the three most voted is equivalent to saying that at least one candidate fears being among the four least voted.

ÿ to question 21 on page 36

Machine Translated by Google

Question 25 - Answer E is correct

Let's rephrase the hypothesis. State that:

The Commissioner he does not believe that the accused of murder did not go to the victim's house…

it is equivalent to saying that:

the commissioner believes that the defendant went to the same in the time frame in which the house and lost his life.

, therefore, also takes into consideration that the defendant may be the murderer.

ÿ to question 27 on page 43

Question 41 - Answer A is correct

If in the Republic of Arraffa it is false that all parliamentarians are honest and competent, then it is true that there is at least one parliamentarian who is not honest or is not competent. ÿ to question 25 on page 40

Question 26 - Answer A is wrong

The answer shifts the property formulated by Kontakerikonta for the Incredible numbers, of infinite quantity, to those finite in number. The professor does not make any hypothesis about this set. ÿ to question 26 on page 42

Question 29 - Answer B is wrong

It could be true; from the conditions of the problem there is no information to exclude that in reality the Tondolini are a subset of the Marziolani, with the same characteristics.

ÿ to question 29 on page 46

Question 13 - Answer E is wrong

The reasoning proposed is true: if Mario were a professor, he would have a register. We must, however, find the statement certainly false. Let's interpret the proposed deductions with mathematical logic:

ÿ is equivalent to (NOT) ÿ (NOT)

professor ÿregister is equivalent to NON register ÿNOT professor.

ÿ to question 13 on page 25

Question 31 - Answer A is wrong

Piero is the second richest.

ÿ to question 31 on page 51

Machine Translated by Google

Question 41

In the Republic of Grab it is NOT true that

Every member of parliament is an honest and competent person

So, in the Republic of Arraffa:

A. some members of parliament are not honest or are not competent ÿ page 66 B. every member of parliament is either honest, but not competent, or is ÿ page honest C. some members of 69 competent, but not parliament are honest and competent ÿ page 59 D. honest and competent people never become members of parliament ÿ page 12 E. no member of parliament is honest and competent ÿ page 12

Question 35 - Answer A is correct

- In sequence we find a triangle, a square, a triangle: the missing figure seems to have to be a square.
- If we agree to think of the numbers written outside the polygon as positive, both on the sides and in the vertices, and those inside as negative, we can identify the law that links them: the number at the vertex is the algebraic sum of the numbers written on the sides, positioned internally if negative, externally if positive, according to the established convention.
  In fact, if we analyze the vertices of the first triangle:

  15 = 9 + 6; 3 = 6 ÿ 3; 6 = ÿ3 + 9

  if we analyze the vertices of the first square:

  28 = 16 + 12; 4 = 12 ÿ 8; ÿ4 = ÿ8 + 4; 20 = 4 + 16

  if we analyze the vertices of the second triangle:

  3 = 9 ÿ 6; ÿ9 = ÿ6 ÿ 3; 6 = ÿ3 + 9

- Only the square in figure 2 enjoys this property

  ÿ4 = ÿ16 + 12; 4 = 12 ÿ 8; ÿ4 = ÿ8 + 4; ÿ12 = 4 ÿ 16

- No other figure, neither square nor hexagon, enjoys this property.

ÿ to question 16 on page 28

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Question 42

A welcoming sign at the entrance to the L'Oca Giuliva restaurant reads:

If you are few, you eat well

If there are many of you, you spend little

Mr. Aquilotto, with his sharp mind, logically deduces that:

A. if there are few of you, you spend a lot B. to eat well you need to go with a few people C. if you eat badly you are not a few people D. to spend little you need to be a lot E. if there are a lot of you, you eat badly

ÿ page 76 ÿ page 51 ÿ page 13 ÿ page 41 ÿ page 34

Question 33 - Answer A is wrong

The property must be equal to ÿ , an integer, therefore also for = 1.
Let ÿ be the set of prime numbers, it follows that if

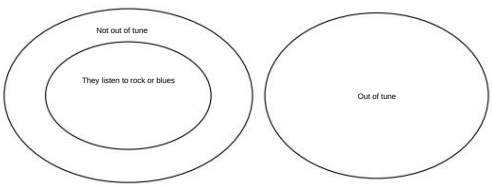
= 1 ÿ ÿ > 1, ÿ ÿ ÿ + 2 ÿ ÿ

Then prime numbers of type + 2 do not exist. The negation of the conjecture does not state that they do not exist, but that they are finite.

ÿ to question 33 on page 53

Question 23 - Answer B is wrong

Let's represent the first hypothesis of the question using Venn diagrams:



We can deduce that a tone deaf person does not listen to rock music.

ÿ to question 23 on page 38

Machine Translated by Google

Question 43

The King did not respect the advice of the Grand Chamberlain to oppose the celebration of the marriage of the Swan-Necked Princess to the toad she loved, if the young people insisted on celebrating the rite in the Basilica of Superga.
Princesses, at least those in fairy tales, follow their father's will. What do you deduce from this?

A. The princess and the toad will be able to get married, if they wish, in the Basilica of Superga ÿ page 42

B. The princess and the toad will not get married ÿ page 50

C. The princess and the toad will be able to marry, but not in the Basilica ÿ of Superga page 16

D. The data of the problem do not authorize us to conclude that any of the proposed interpretations are true. ÿ page 53

E. The princess and the toad will necessarily marry in the Basilica of Superga ÿ page 43

Question 41 - Answer B is wrong

From the hypothesis

It is not true that every MP is an honest and competent person, we deduce that there can be MPs in Arraffa who are both honest and competent, neither honest nor competent, honest but not competent, not honest but competent. Surely, however, there is at least one MP who does not possess at least one of the two qualities!
Honesty does not preclude competence, and competence does not preclude honesty.

ÿ to question 41 on page 67

Question 46 - Answer C is wrong

Give an example and you will see that the right answer is another.

ÿ to question 46 on page 73

Question 4 - Answer A is wrong

There is only one correct answer for every well-formulated question, but this is not the correct answer. The law of formation of pairs of numbers has not been identified. In fact, what relationship does 40 have with the sequence of the first numbers of the previous pairs, and 80 with the sequence of the second numbers of the previous pairs?

ÿ to question 4 on page 10

Machine Translated by Google

Question 25 - Answer C is wrong

The commissioner believes that the defendant went to the house where he lost his life. in the time interval in

However, from the hypotheses we have no information to know whether the commissioner considers it strangler or not.

ÿ to question 25 on page 40

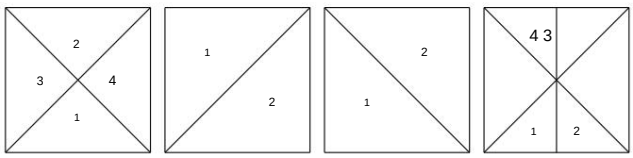
Question 44 - Answer B is wrong

There are 30 first-term parliamentarians; the parliamentarians investigated and definitively condemned there are 40. We have no information on whether any of the 30 first-time parliamentarians are both in the group of 40.

ÿ to question 44 on page 71

Question 2 - Answer B is correct

There are 12 triangles:



4 + 2 + 2 + 4 = 12

ÿ to question 51 on page 79

Question 49 - Answer E is wrong

Umberto could have no children or have children, for example, with black hair.

ÿ to question 49 on page 76

Question 4 - Answer B is wrong

The answer to a question is always unique: unfortunately there is another correct answer! In fact, that relationship has 46 with the sequence of the first numbers of the previous pairs, and 64 with the sequence of the second numbers of the previous pairs?

ÿ to question 4 on page 10

Machine Translated by Google

Question 44

Of the 120 parliamentarians of Allegrandia, it is known that a third have been investigated by the judiciary and definitively condemned and three quarters are in second (or in any case, not in first) parliamentary mandate. It can be concluded that:

A. a quarter of the parliamentarians are in their first term and have been definitively sentenced ÿ page 73

B. none of the first-term parliamentarians have been convicted definitely ÿ page 70

C. three parliamentarians were chosen anyway, at least one of them was definitively sentenced ÿ page 49

D. a third of first-term MPs have been convicted definitely ÿ page 29

And there is at least one member of parliament who has been definitively convicted and it is a mandate after the first ÿ page 62

Question 33 - Answer D is correct

To disprove the conjecture means to prove that the negation of the conjecture is true. Let's rewrite the conjecture:
Prime numbers, such that + 2 is a prime number, are infinite
The negation of the conjecture becomes:
Prime numbers, such that + 2 is a prime number, are finite in number
that is, taking into consideration increasingly larger positive numbers, from a certain moment in then we no longer encounter prime numbers that satisfy the property of the conjecture.
The negation of the conjecture can be rewritten by replacing from a certain point onwards… with:
there exists an integer such that whatever the prime number with > e + 2 is not prime.
The conjecture we have been discussing is the famous twin prime conjecture: prime numbers that they are 2 units apart.

Examples of twin primes are:

(3,5), (5,7), (11,13), … , (239,241), … , (347,349), (461,463), …

but also

694 513 810 ÿ 22304 ± 1, 1 159 142 985 ÿ 22304 ± 1

both of 703 digits.

ÿ to question 24 on page 39

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Question 45

Indicate which of the pairs of numbers indicated should be inserted in place of the dots in the following sequence

3, 43 ; 5, 27; 9, 19; …, …; 33, 13

A. 24, 74
B. 19, 11
C. 15, 15
D. 17, 15
E. 23, 13

ÿ page 76 ÿ page 45 ÿ page 51 ÿ page 57 ÿ page 49

Question 4 - Answer C is wrong

There is only one correct answer, for every well-formulated question! In fact, what relation does 40 have with the sequence of the first numbers of the previous pairs, and 67 with the sequence of the second numbers of the previous pairs?

ÿ to question 4 on page 10

Question 2 - Answer A is wrong

There are more triangles.

ÿ to question 2 on page 8

Question 37 - Answer A is correct

Let's rewrite the hypotheses:

if the budget is not cut: prices will remain stable if and only if all taxes are increased

The budget was not cut, all taxes were increased, so prices remained stable.
With the laws of mathematical logic:

ÿ

is equivalent to

ÿ and ÿ

stable prices ÿall taxes have been increased all taxes have been increased ÿstable prices

of course with an uncut budget.

ÿ to question 39 on page 64

Machine Translated by Google

Question 46

The great number theorist Valakekontojioo, studying the integers 1,2,3,4,5, … found that among them there could exist the Cyrillic numbers, which enjoy these two properties:

- the sum of two Cyril numbers (even equal ones) is a Cyril
- the product of two Cyril numbers (even equal ones) is not a Cyril

His student Son Pyooh Foorb carefully studied these numbers, and discovered how many Cyrillic numbers there are, and precisely he deduced that the number of Cyrillic numbers is:

A. 3
B. 0
C. 1
D. 4
E. infinity

ÿ page 18 ÿ page 64 ÿ page 69 ÿ page 7 ÿ page 41

Question 16 - Answer E is wrong

It is true that the total combinations of the five signs + and ÿ are 2 32, but many combinations of signs do not lead to different results.

ÿ to question 16 on page 28

Question 16 - Answer C is wrong

Every number has its opposite (try to write some) and zero cannot be obtained. So from algebraic sums of the type indicated you can only get an even number of cases.

ÿ to question 16 on page 28

Question 44 - Answer A is wrong

The 30 first-term MPs (a quarter of the MPs) could all be virtuous or NOT have been definitively convicted. In fact, 40 out of 120 have been definitively convicted and they could all be MPs with a mandate after the first (80 out of 120). ÿ to question 44 on page 71

Question 18 - Answer C is wrong

We have no information to deduce that there were also two cups of coffee on the table.

ÿ to question 18 on page 33

Machine Translated by Google

Question 47

Archaebacteria are unicellular organisms that live in ponds and reproduce by fission (that is, each bacterium divides in two and forms two other identical bacteria).
If you put an archaebacterium in a certain pond, it reproduces by fission only once each day, and after 30 days the surface of the pond is completely covered by the bacteria.

How many days will it take for the surface of the same pond to be completely covered if two archaea are initially introduced?

A. 30 days B. 28 days C. 29 days D. depends on the surface area of the pond E. 15 days

ÿ page 21 ÿ page 74 ÿ page 56 ÿ page 44 ÿ page 10

Question 26 - Answer C is wrong

The answer formulation is incorrect because it does not provide information about the factorization of the Incredible numbers, none of which must have 8 distinct prime factors. This information is essential to prove the falsity of the Kontakerikonta conjecture.

ÿ to question 26 on page 42

Question 6 - Answer C is wrong

Romulus would be sincere and Remus a liar: we know that only one of the two is sincere but we do not know who. Remus could be the sincere brother.

ÿ to question 6 on page 12

Question 32 - Answer B is wrong

All and only engineers are trustworthy people; some trustworthy people deal with politics, therefore some engineers deal with politics.

ÿ to question 32 on page 52

Question 47 - Answer B is wrong

28 days is too short, because only half of the pond is covered.

ÿ to question 47 on page 74

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Question 48

In a discussion between friends, Antonio says:

We all like coffee, except Paola, who doesn't like it.

Fabio observes that Antonio is wrong.

It follows that:

A. all the friends like coffee B. Paola ÿ page 80
likes coffee C. one of the ÿ page 20
friends, who is not Paola, doesn't like coffee D. either Paola ÿ page 21
likes coffee, or there is someone among the friends,
besides Paola, who doesn't like coffee ÿ page 56
And it is not possible that only one of the friends dislikes the coffee ÿ page 48

Question 28 - Answer E is correct

Let's count the number of digits needed to number the pages of the book.

to number the pages between…	necessary digits	total digits
1 ÷ 9	1 ÿ 9	9
10 ÷ 99	2 ÿ 90	180
100 ÷ 999	3 ÿ 900	2700
1000 ÷ 1500	4 ÿ 501	2004

So the number of digits to number the pages of a book is:

number of pages	figures needed	total pages	
999	9 + 180 + 2700	2889	few!
1500	+ 2004	4893	too many!

The pages are between 1000 and 1500.

ÿ to question 35 on page 55

Machine Translated by Google

Question 49

Indicate what the negation of the statement is

Umberto has at least one blond son

A. At least one of Umberto's children is not blond B. Umberto has no children or only non-blond children C. All of Umberto's children are brunette D. Not all of Umberto's children are blond E. Umberto has all his children with red hair

ÿ page 20 ÿ page 60 ÿ page 29 ÿ page 33 ÿ page 70

Question 42 - Answer A is wrong

The hypothesis

if there are many of you, you spend

little it does not provide information on the expenditure (per person) if the dinner is with a few friends.

With the laws of mathematical logic, I place

= there are many of us and = we spend little

we know that

ÿ NOT equivalent to (NOT) ÿ (NOT)

So, if there are NOT many of you, that is, there are few of you, how much do you spend (per person)?

Mr. Aquilotto has no information to deduce that one is spending NOT little but a lot.

ÿ to question 42 on page 68

Question 1 - Answer A is correct

The delivery is made with a van, so Aldo or Ciro must be driving. If Ciro is there, Aldo must also be there (be careful because Aldo could make the deliveries without Ciro): so Aldo is certainly present. We cannot know if Ciro or Bruno is there, but Aldo is certainly paid, because he is present.

ÿ to question 50 on page 77

Question 45 - Answer A is wrong

Neither number is exact. Reflect and look for another law of formation of pairs of numbers. The first number of each pair belongs to an increasing sequence, while the second to a decreasing one.

ÿ to question 45 on page 72

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Question 50

Complete the following sequence with the two appropriate numbers:

7 14 29 60 123

A. = 63, B. = 32 ÿ page 44
= 250, C. = = 500 ÿ page 11
250, D. = = 505 ÿ page 14
224, E. = = 283 ÿ page 17
240, = 485 ÿ page 19

Question 39 - Answer E is wrong

If Buffon had saved 12 penalties, Totti would have given him 600 euros.

ÿ to question 39 on page 64

Question 51 - Answer B is correct

To win, the only strategy might be to leave the second player 4 pebbles
on the table, but… to leave 4 the first player would have to take 4 and this is prevented by the rule.
There is nothing left to do but resign ourselves: the first player, no matter what move he makes, loses.

ÿ to question 21 on page 36

Question 6 - Answer B is wrong

In this case both Romulus and Remus would be sincere!

ÿ to question 6 on page 12

Question 14 - Answer C is wrong

The length of the stick, i.e. 90 cm, exceeds one of the three dimensions indicated
normally, 50 cm, 30 cm or 25 cm, so if you want to carry the stick in question with you
you have to pay an additional ticket.

ÿ to question 14 on page 26

Question 28 - Answer C is wrong

The book is too thin; it would have less than 270 pages!

ÿ to question 28 on page 44

Machine Translated by Google

Question 16 - Answer D is correct

- Every number is the algebraic sum of two even numbers (±2, ±4) and three odd numbers (±1, ±3, ±5), therefore it is an odd number;
- every number has its opposite;
- the largest of the numbers is $_1$ = 1 + 2 + 3 + 4 + 5 = 15 which corresponds to the smallest of the numbers $_2$ = ÿ15;
- we verify that we can obtain all odd numbers between ÿ15 and 15.

a) 15 = +1 + 2 + 3 + 4 + 5 b) 13

= ÿ1 + 2 + 3 + 4 + 5 c) 11 = +1

ÿ 2 + 3 + 4 + 5 d) 9 = +1 + 2 ÿ

3 + 4 + 5 or 9 = ÿ1 ÿ 2 + 3 + 4 + 5 e) 7 = +1 + 2 + 3 ÿ 4 + 5 or 7 =

ÿ1 + 2 ÿ 3 + 4 + 5 f) 5 = ÿ1 + 2 + 3 + 4 ÿ 5 or 5 = +1 ÿ 2 ÿ 3 + 4 +

5… g) 3 = ÿ1 + 2 + 3 + 4 ÿ 5 or 3 = ÿ1 ÿ 2 ÿ 3 + 4 + 5… h) 1 = +1 ÿ 2

+ 3 + 4 ÿ 5 or 1 = ÿ1 + 2 ÿ 3 ÿ 4 + 5…

In total there are 16 different numbers, all the odd ones between -15 and 15, 8 positive and 8 negative.

ÿ to question 14 on page 26

Question 25 - Answer D is wrong

Let's rephrase the hypothesis. Say that: he does not believe that the defendant did not go to the victim's house...
it is equivalent to
saying that: the commissioner believes that the defendant went to his home during that period of time.
From the information given, however, we cannot deduce whether the commissioner knows whether the accused is also the strangler!

ÿ to question 25 on page 40

Question 22 - Answer D is wrong

The survey carried out in Sparagna involved 1000 inhabitants aged over 55. We do not know how many, under 55, have a television.

ÿ to question 22 on page 37

Machine Translated by Google

Question 51

Two players take turns taking pebbles with the only rule that they cannot take 4 or 8.

The player who manages to collect the last pebble wins.
If there are initially 8 pebbles, how many must the first player take to ensure victory, assuming that in subsequent moves each player makes no mistakes?

A. Whatever number he picks, he will always win B. Whatever number he picks, he will always lose C. 1

D. 2

E. 3

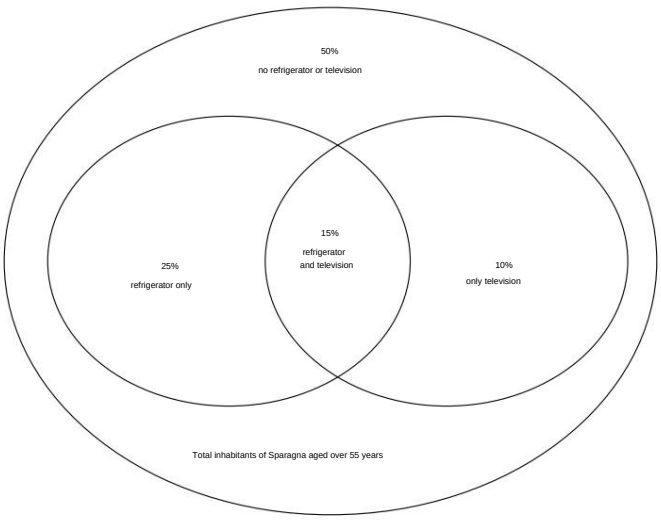
ÿ page 55 ÿ page 77 ÿ page 43 ÿ page 32 ÿ page 55

Question 22 - Answer C is correct

Since in Sparagna the citizens over 55 years of age who have a television or a refrigerator or both are half:

400 + 250 ÿ 150 = 500 ÿ 50%

the other half of the population, that is 50%, has neither a television nor a refrigerator.
Let's visualize the situation with a Venn diagram scheme.



ÿ to question 11 on page 22

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Question 48 - Answer A is wrong

The denial of the affirmation

We all like coffee, except Paola, who doesn't like it.

And

Either Paola likes coffee or there is someone among her friends, besides Paola, who doesn't like coffee

The proposed answer only considers the case in which Paola also likes coffee and not the possibility that there is some other friend who does not like it, since Fabio claims that Antonio is wrong.

ÿ to question 48 on page 75

Question 17 - Answer B is wrong

3 becomes 5 by turning one more bar on the display and turning another off.

ÿ to question 17 on page 30

Question 30 - Answer C is correct

We are given a set of 5 integers, three of which are odd and two are even. In the first position we can put any of the three odd numbers (29, 43, 55); in the second position, for each of the three odd numbers, we can choose between the two even numbers (28, 36). The cases then become

3 ÿ 2 = 6

In third position, for each of the previous six cases, we still have two choices between the two remaining odd numbers. The possible cases become

3 ÿ 2 ÿ 2 = 12

For the fourth and fifth positions we have NO more choices, but we must complete each sequence with the only remaining even and the only remaining odd.

ÿ to question 4 on page 10

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Reading the mentor booklet is finished.